Evolution of Cosmic Structure

Lecture 13 - Observations of distant galaxies

Suhail Dhawan

UNIVERSITY OF BIRMINGHAM

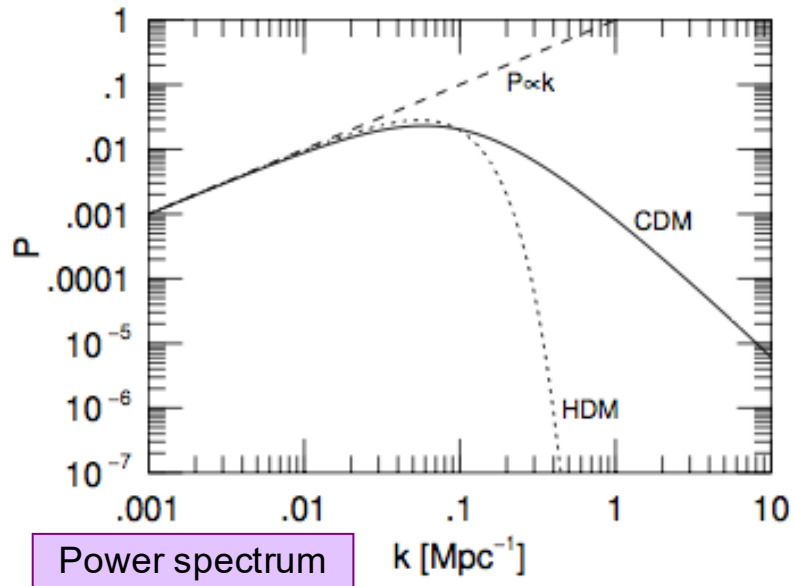
s.dhawan@bham.ac.uk

Learning Outcomes

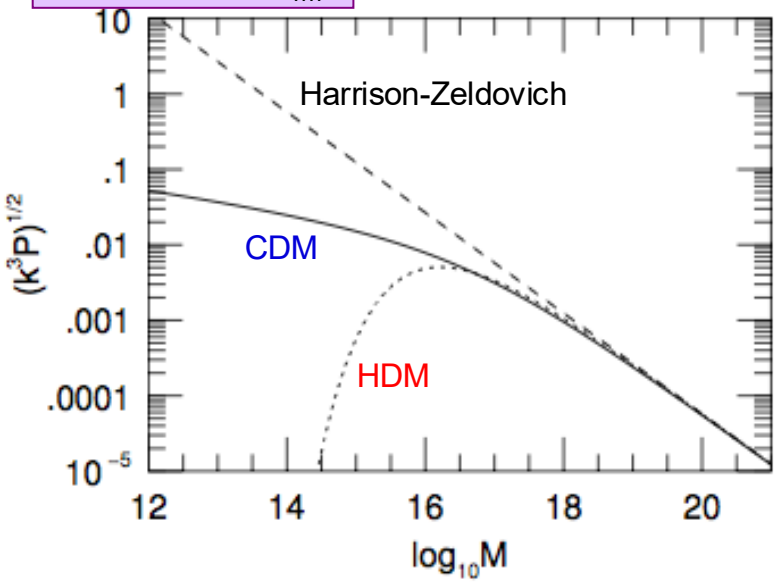
What do we know of the first galaxies and the history of cosmic star formation? What effect has the development of galaxies had on the rest of the baryonic matter?

- Observations of high redshift galaxies
- Reionization – the impact of star formation on the intergalactic medium
- The history of star formation and the growth of stellar mass in the Universe

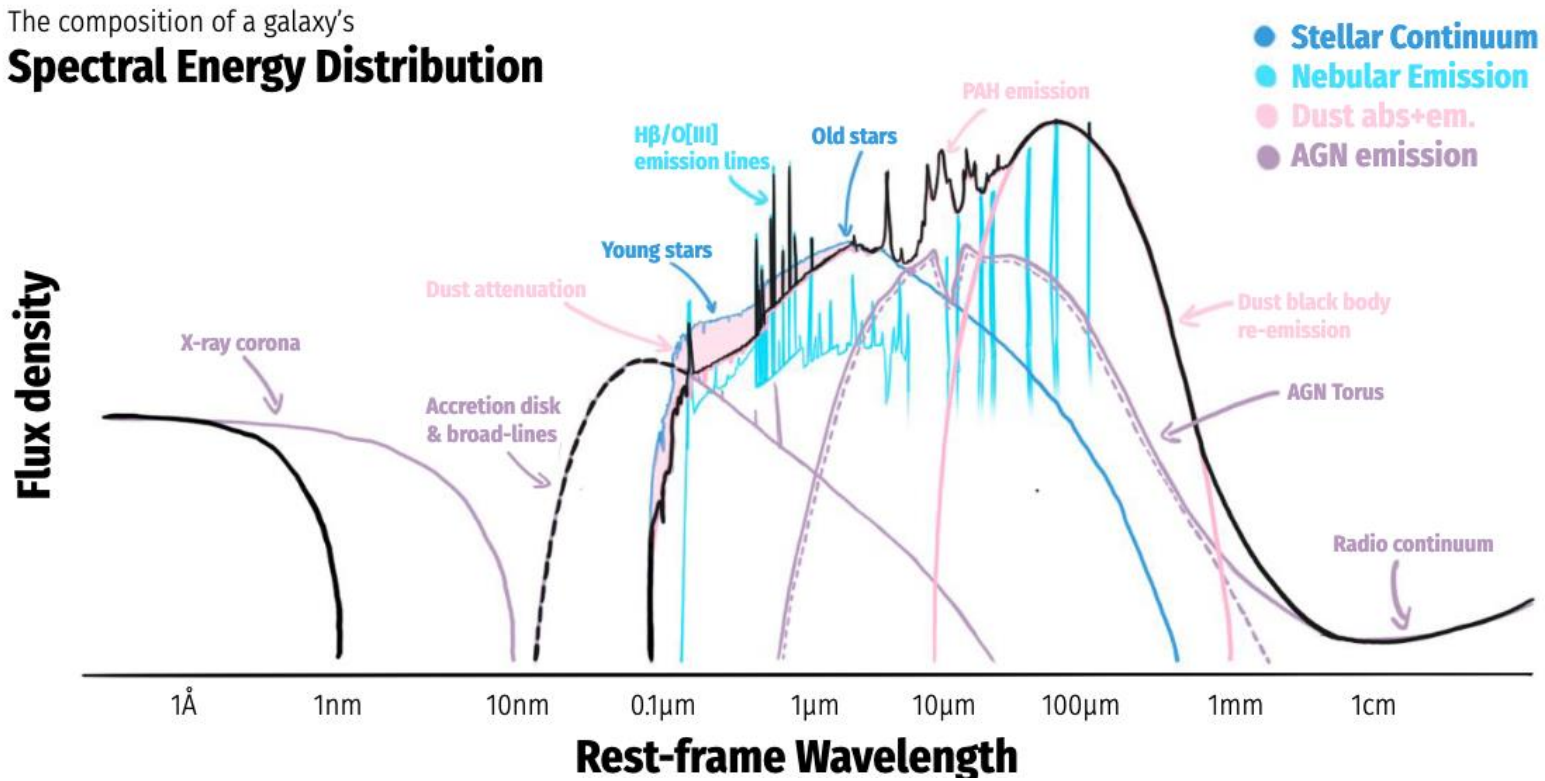
Why do we want to look at high z galaxies?



Power spectrum and $\Delta M/M$ at t_{rm}

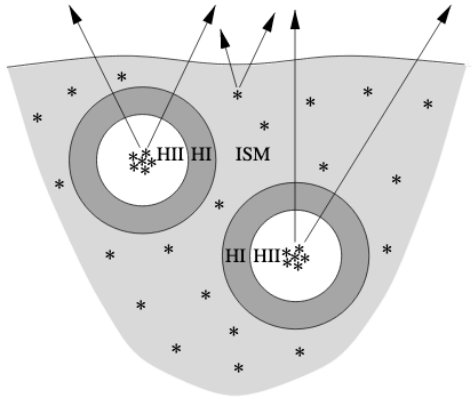


The spectral energy distribution of galaxies



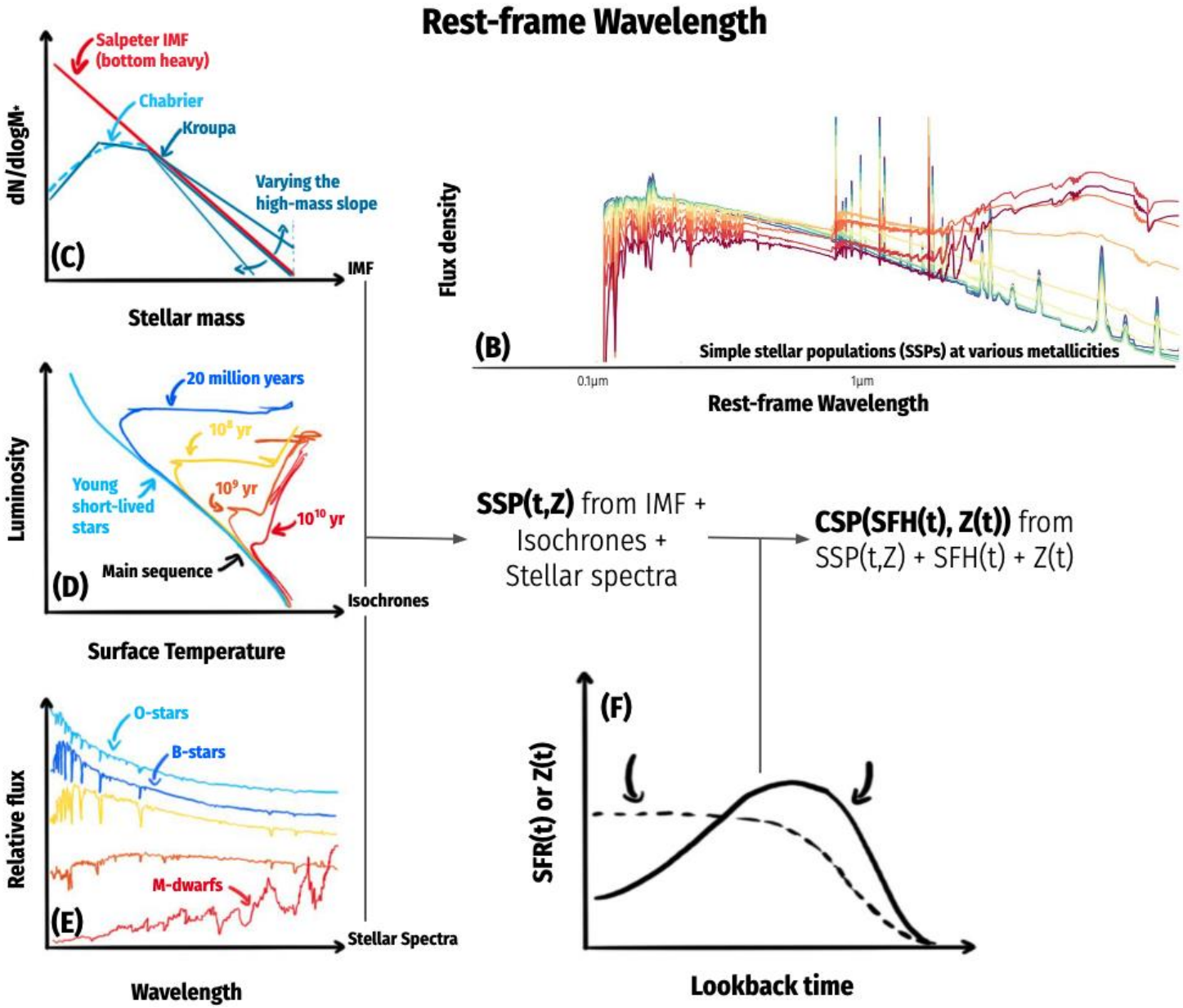
(Iyer et al. 2025)

`normal' star-forming galaxy
-unattenuated stellar spectrum
-emission by dust in stellar birth clouds
-emission by dust in the ambient interstellar medium
-total resulting emission from the galaxy



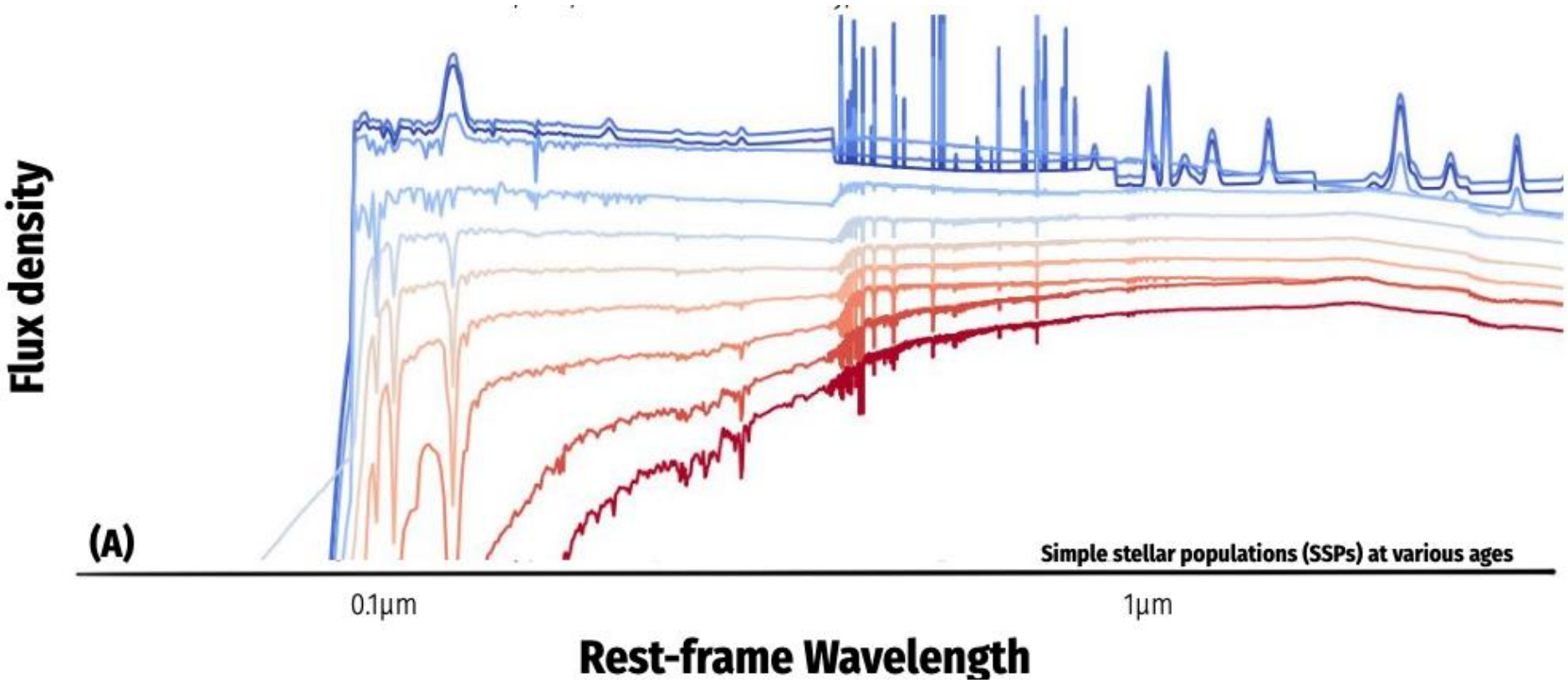
The spectral energy distribution of galaxies

Need to exploit the SED of galaxies to find high redshift ones.



(Iyer et al. 2025)

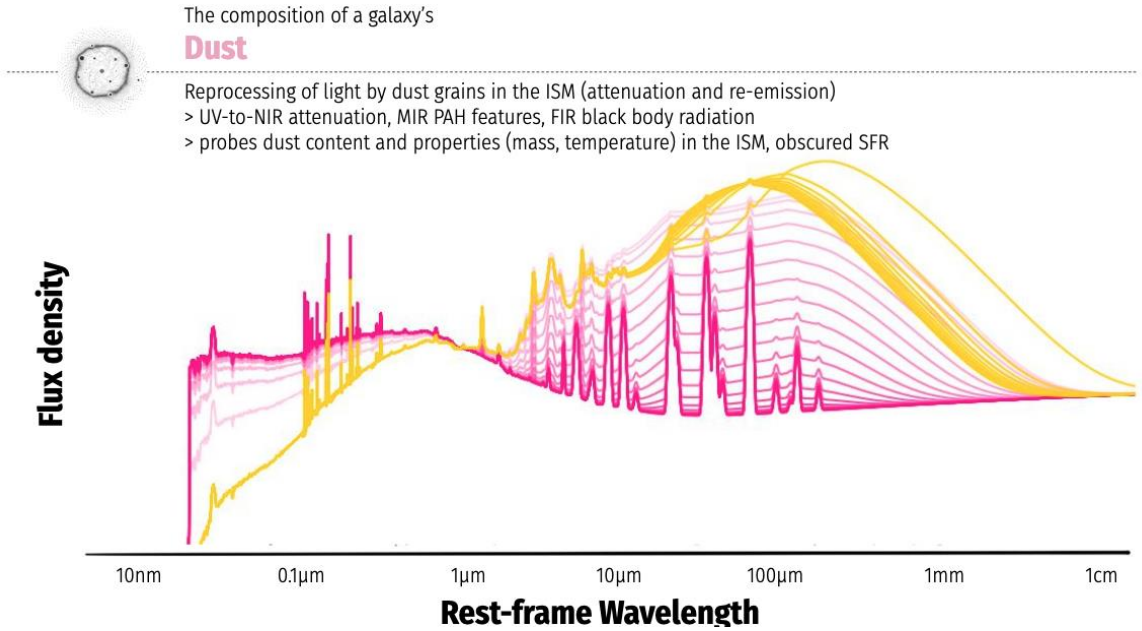
The spectral energy distribution of galaxies



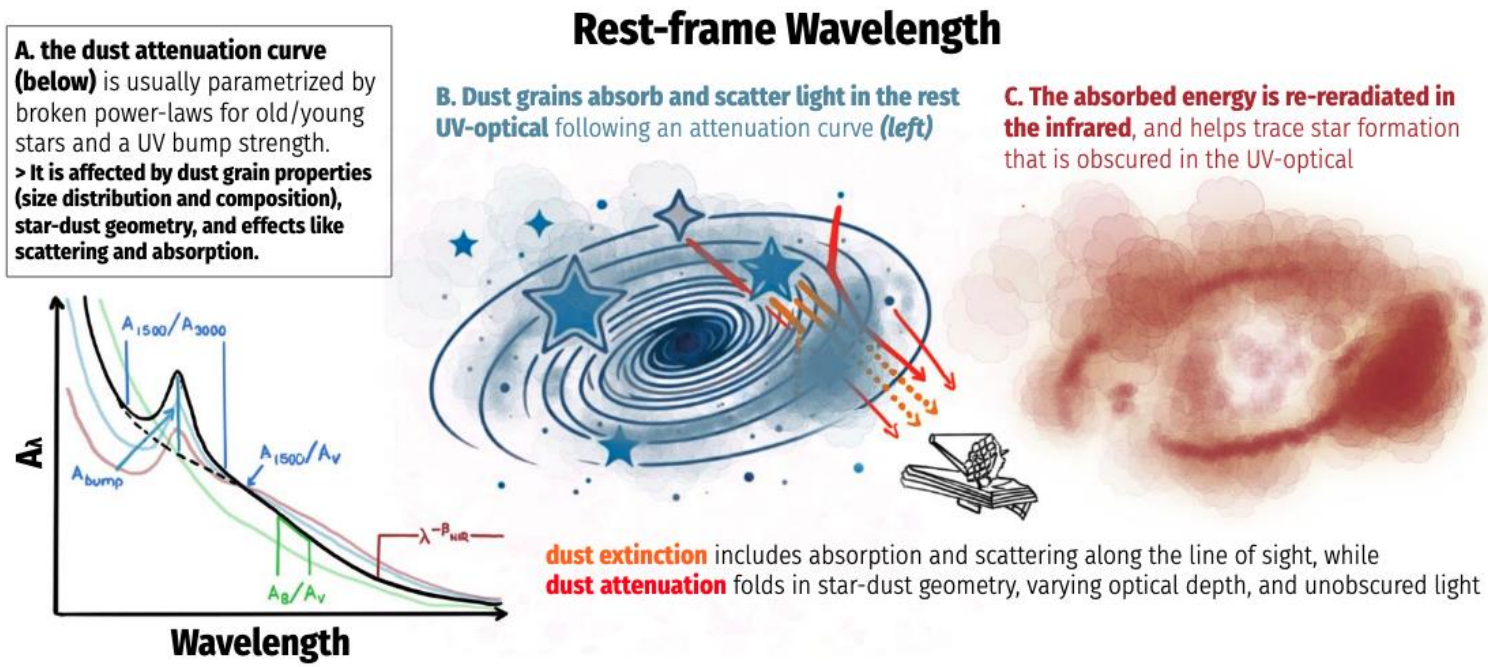
Need to exploit the SED of galaxies to find high redshift ones.

(Iyer et al. 2025)

The spectral energy distribution of galaxies



Need to exploit the SED of galaxies to find high redshift ones.



(Iyer et al. 2025)

Detecting high z galaxies

High redshift galaxies can be detected using a variety of methods:

- **UV dropout techniques** – were pioneered by Steidel and coworkers. The Lyman break (sharp drop in flux blueward of the Lyman limit, at 912 Å) can couple with a suitable choice of filters to produce sources which "disappear" from one band to another.

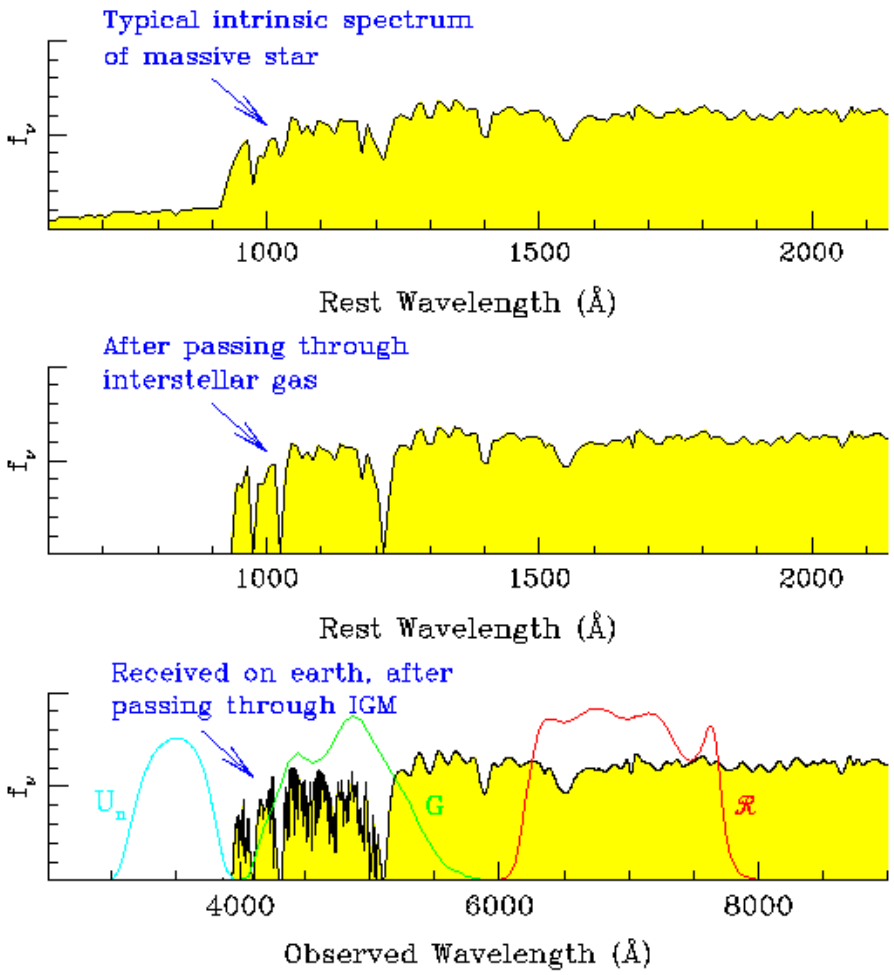
- **Other dropout techniques** – the same idea can be adapted to find galaxies at other redshifts, using different filters and the Lyman or Balmer (3646 Å) break.

- **Submm detection** – the strong far-IR spectral peak at λ~50 µm in star-forming galaxies (due to reradiation of UV by warm dust) leads to very strong submm flux from high *z* galaxies.

- **Narrow band imaging** – picking out objects with a strong Ly α line. This tends to pick up lower luminosity star forming galaxies than the above techniques, but they have faint continua, and are hard to follow up.

- **Direct spectroscopy**

Lyman break galaxies



Use some characteristic ‘feature’ in the spectrum of the galaxy. In this case, the Lyman break which occurs at a rest frame wavelength of 912 angstroms.

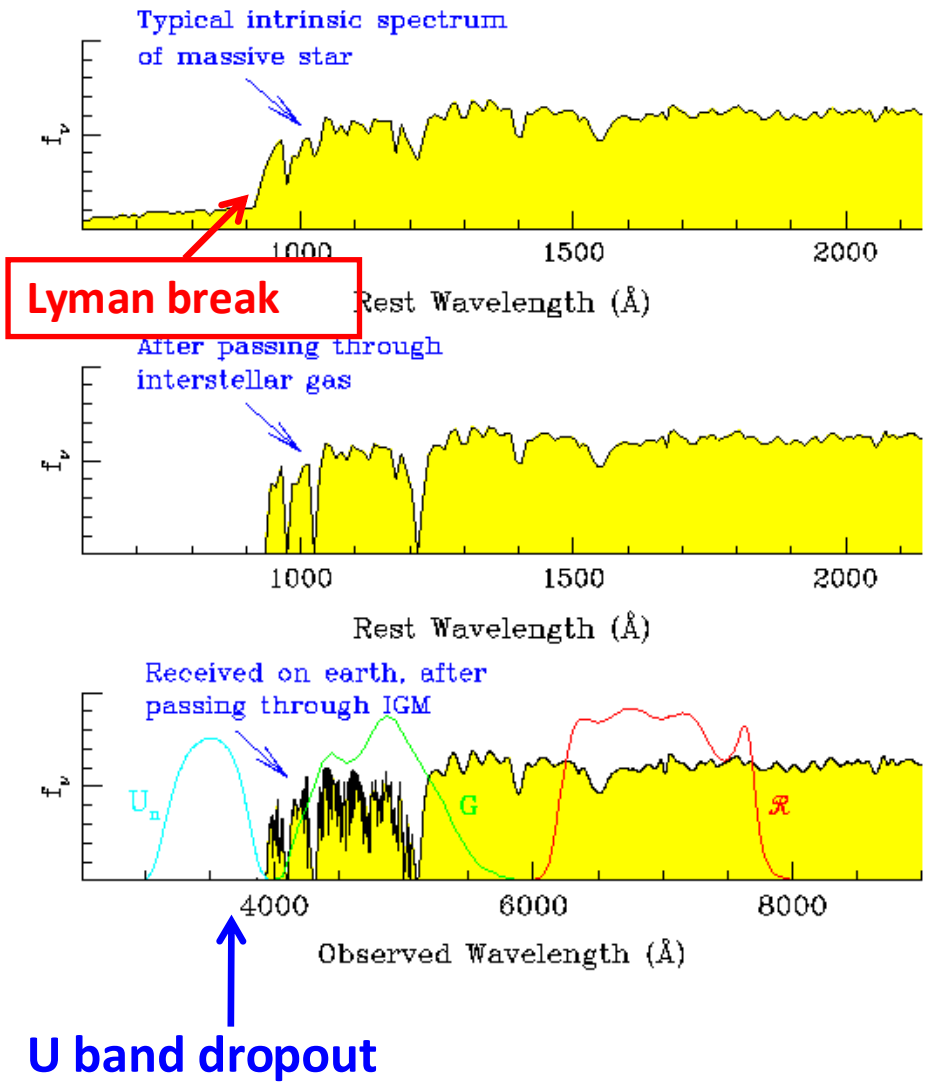
This Lyman break feature is in the minimum energy to ionize a hydrogen atom in the n = 1 state.

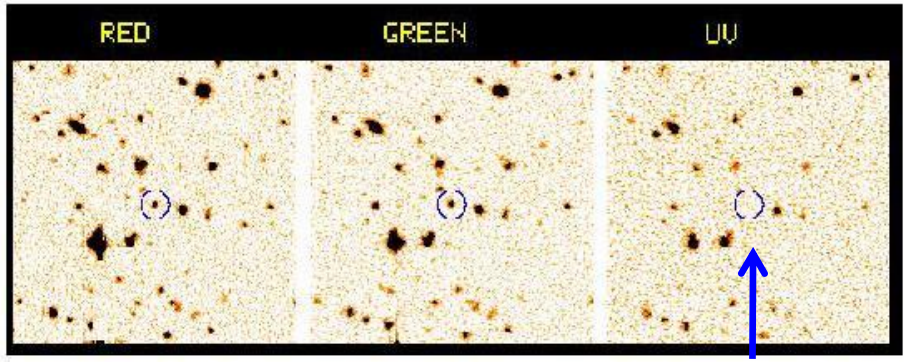
Observed wavelength = (1 + z) * emitted wavelength

More absorption from the interstellar and IGM makes Even clearer feature.

Similar features exist at Balmer break (3646 Angstroms)

Lyman break galaxies





U band dropout

The Lyman continuum discontinuity is particularly powerful for isolating star-forming high redshift galaxies.

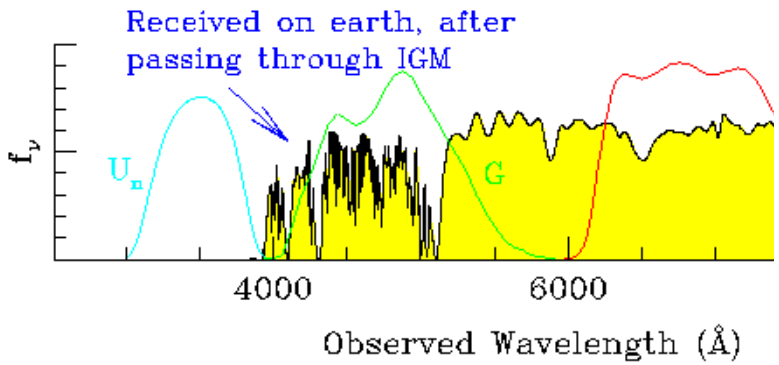
From the ground, we have access to the redshift range z=2.5-6 in the 0.3-1 micron range.

Steidel et al 1999, Adelberger et al 2004

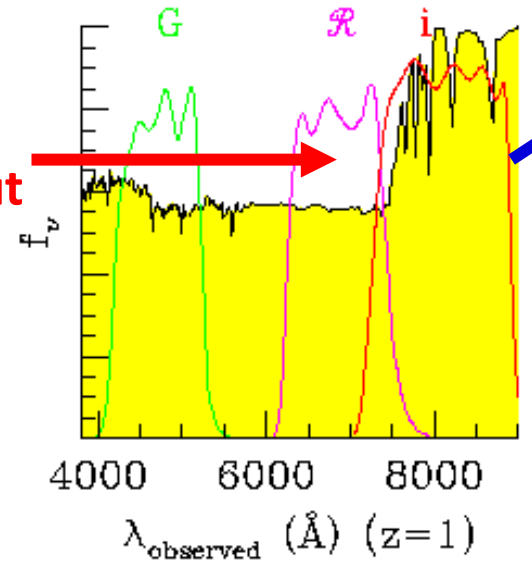
Other dropout galaxies

Extending the technique to 1 < z < 4

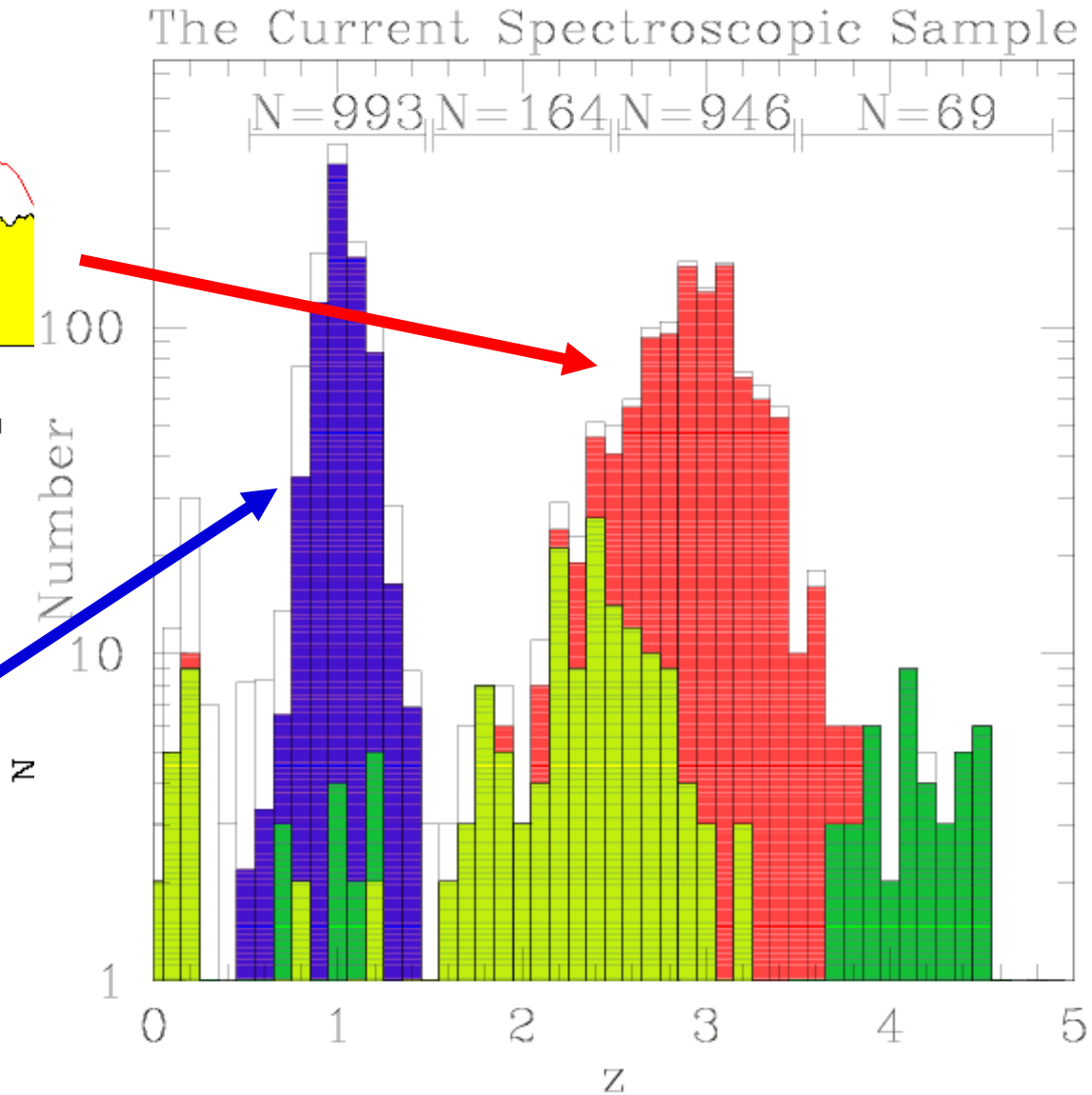
Lyman break



Balmer break



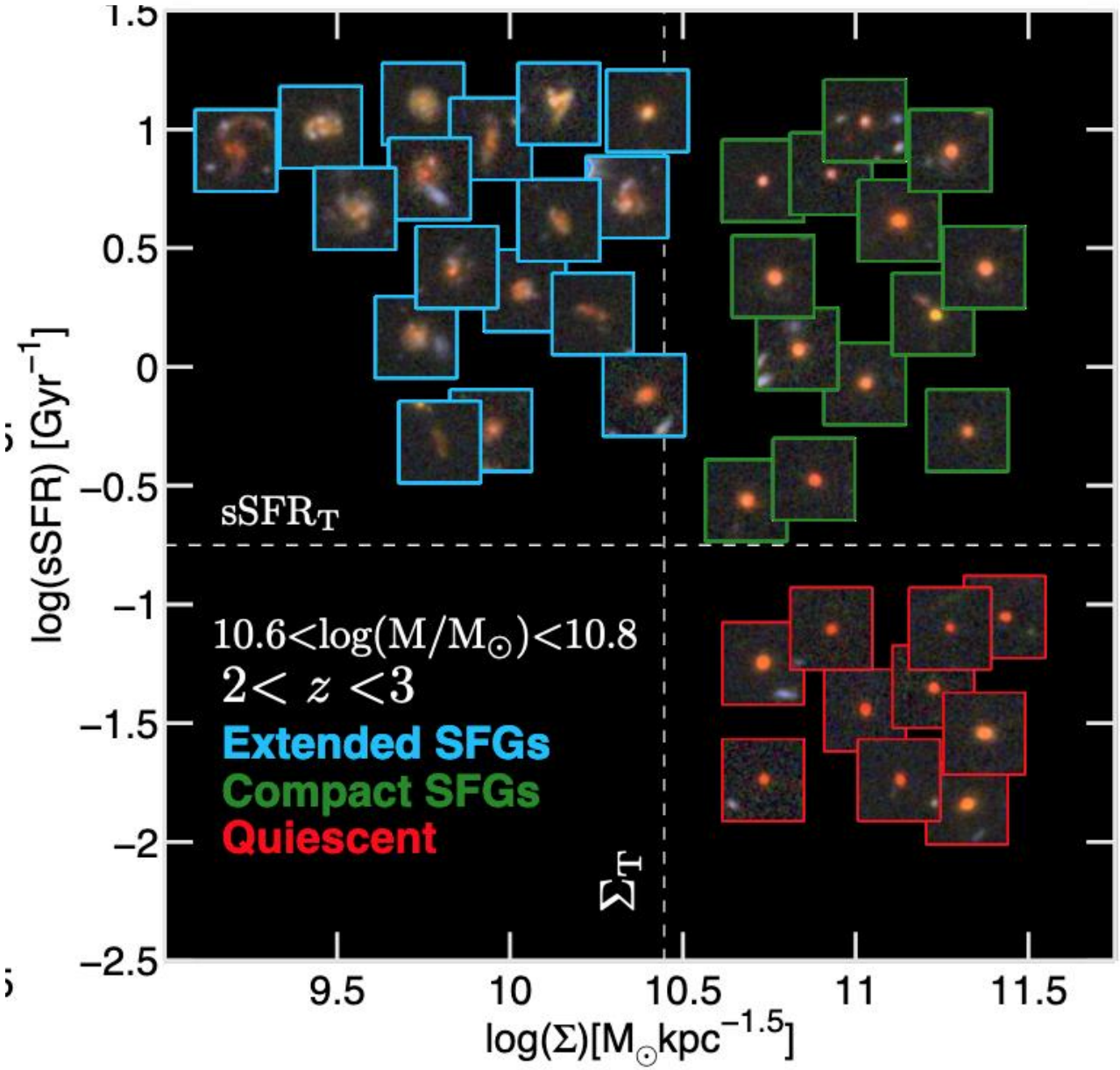
R band dropout



HST morphologies

HST images of spectroscopically-confirmed "Lyman break" galaxies with z>2 in Hubble Deep Field North revealing small physical scale-lengths and irregular morphologies

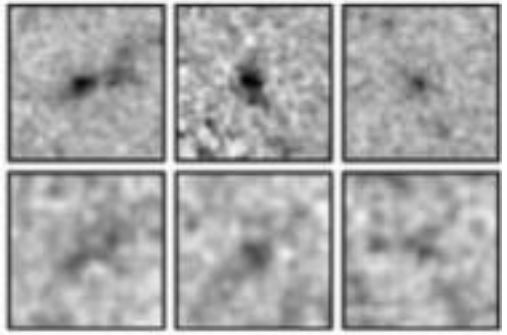
LBGs typically have stellar masses of a few$\times10^{10}M_\odot$ and SFRs of 10-100 $M_\odot$ yr^{-1}.



Barro et al. 2014, 2016

High *z* galaxies

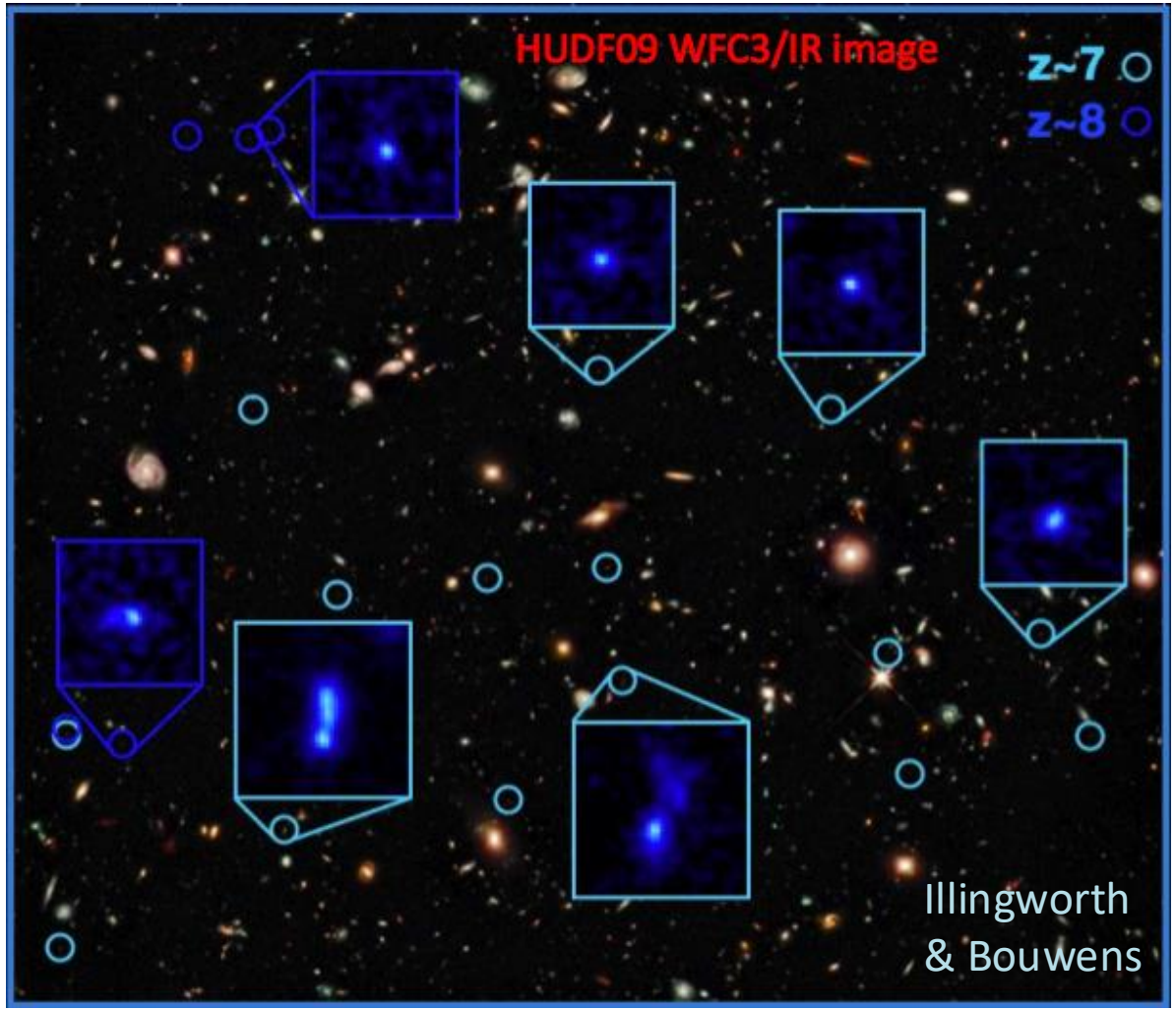
Prior to JWST, application of dropout techniques to the Hubble Ultra-Deep Field, combining HST (~1μm) and Spitzer (~4 μm) data, have pushed the detection of galaxies out to *z*~8. The new WFC3 is one of the keys to this development.



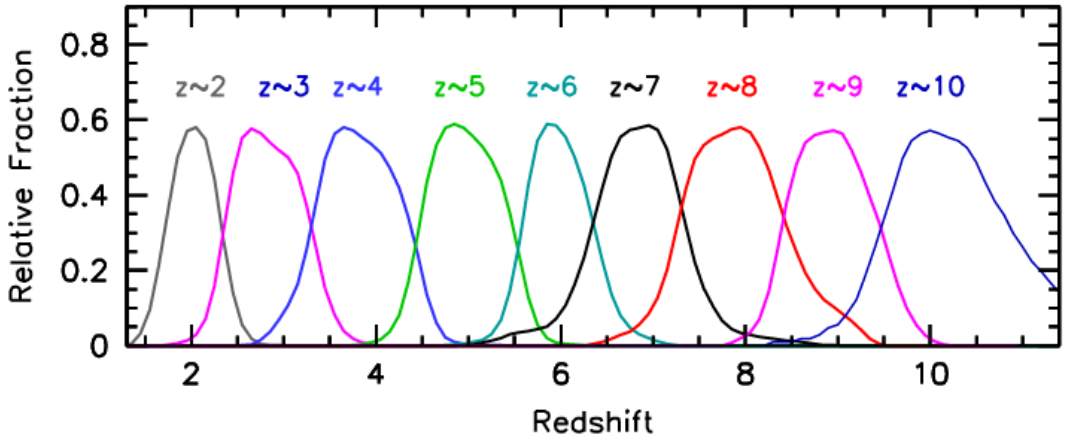
WFC3/IR

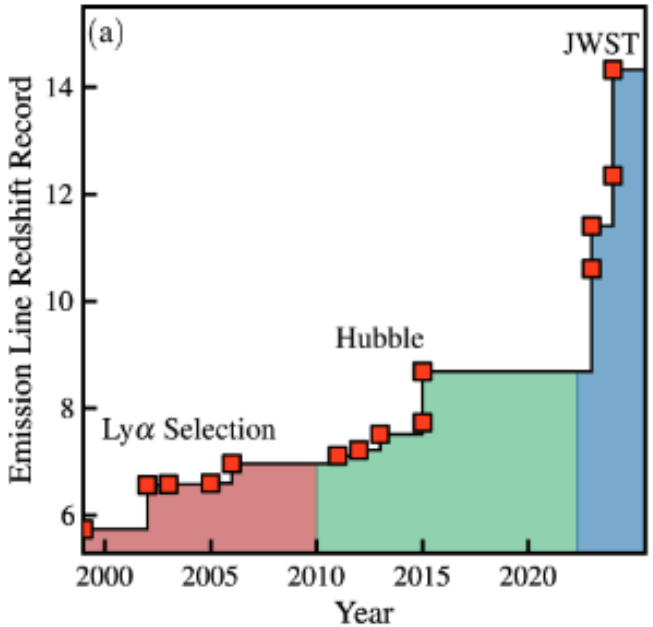
NICMOS

These galaxies are small (typically <10% of the mass of the Milky Way), and are forming stars only ~650 Myr after the Big Bang.

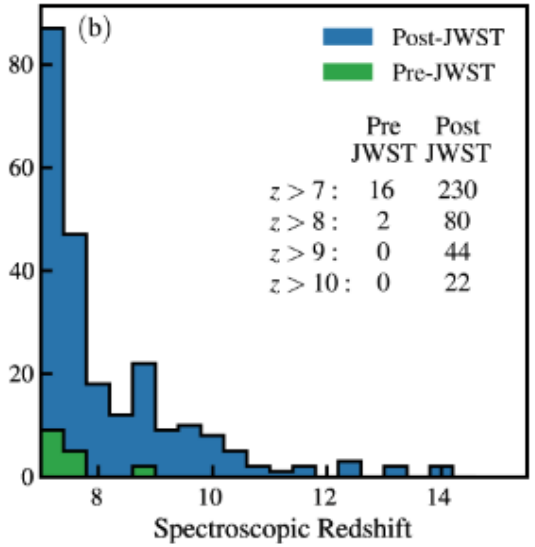


High *z* galaxies



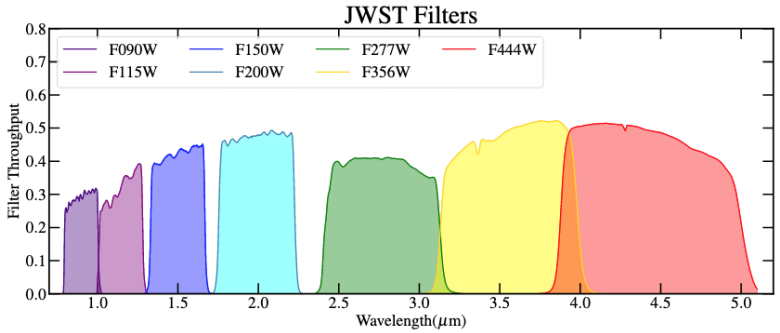


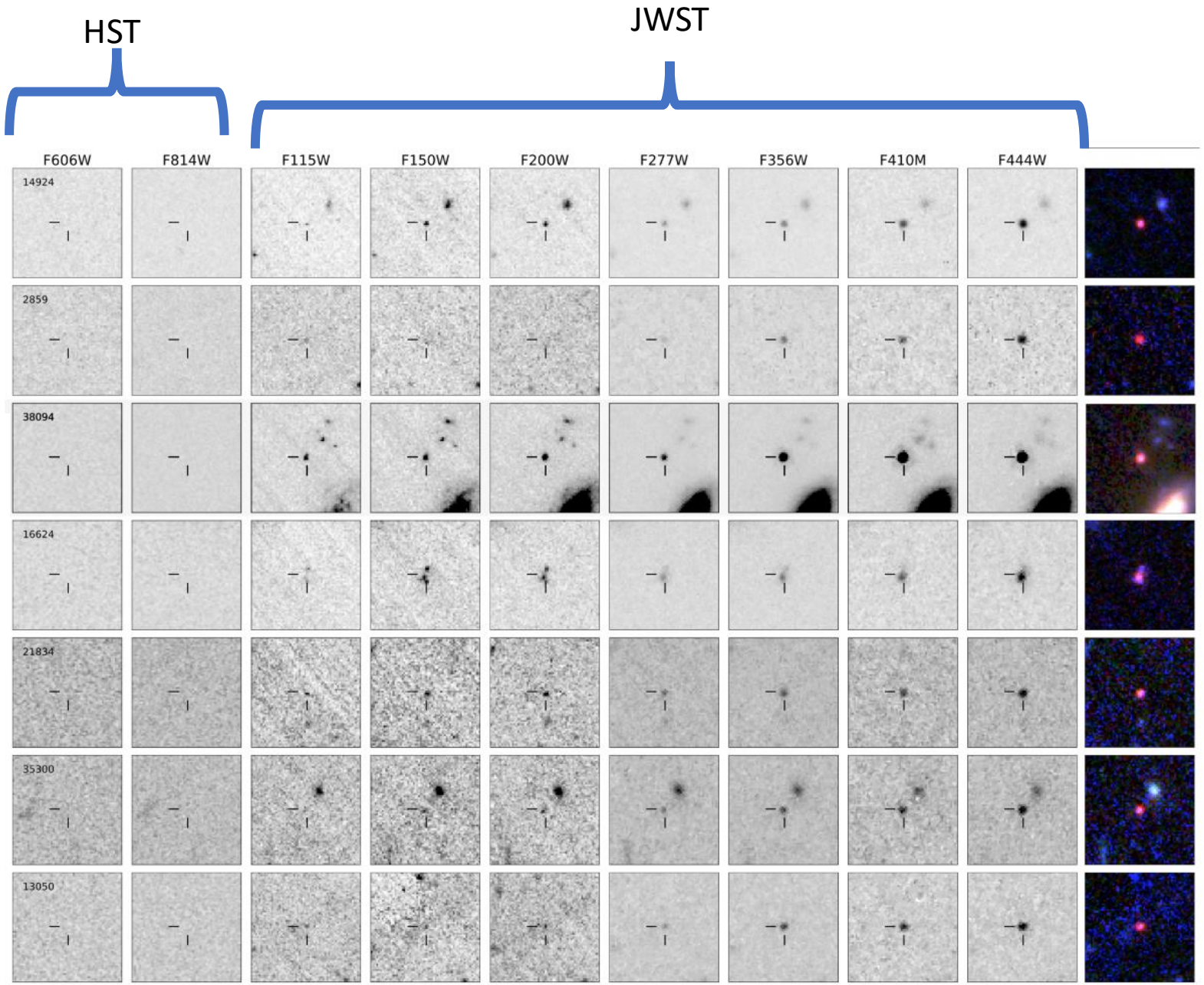
Can now measure large samples across many redshifts self-consistently



Bouwens et al 2021

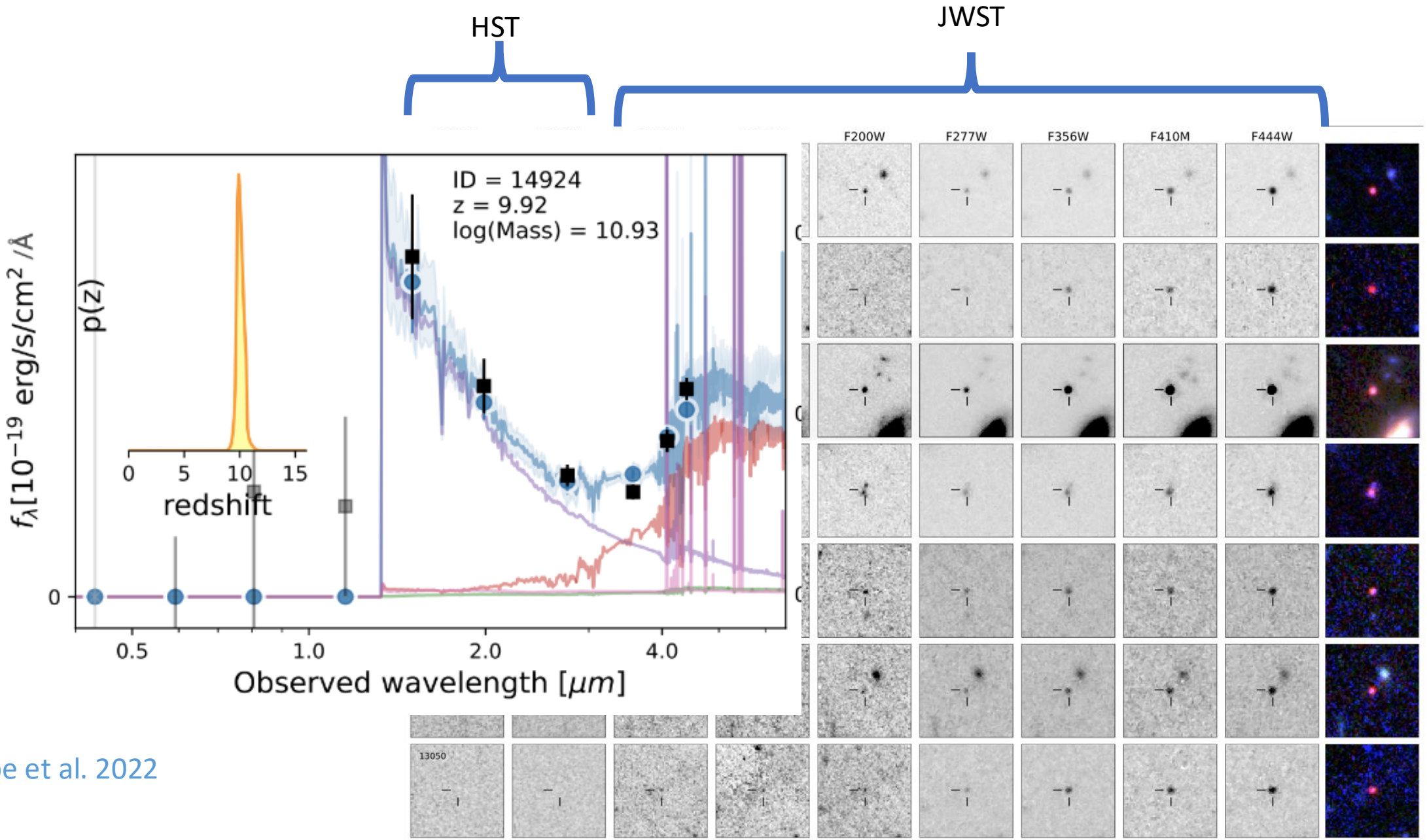
High redshift galaxies (8 < z < 20)





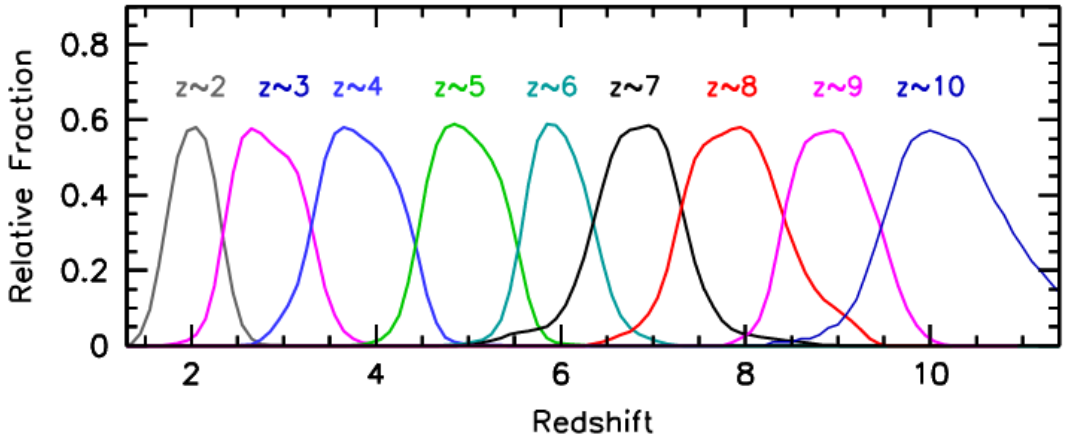
Labbe et al. 2022

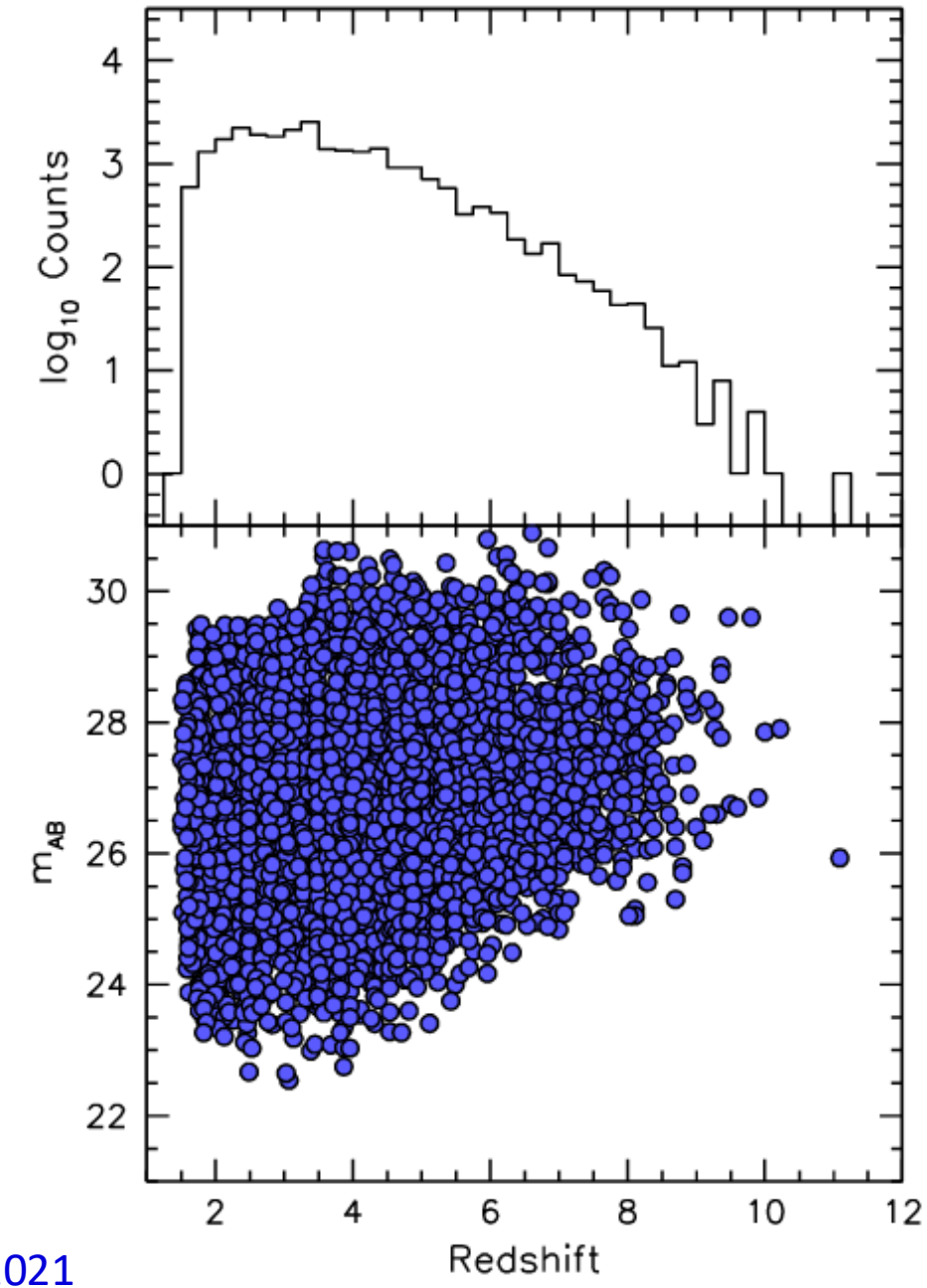
High redshift galaxies (8 < z < 20)



Labbe et al. 2022

High *z* galaxies

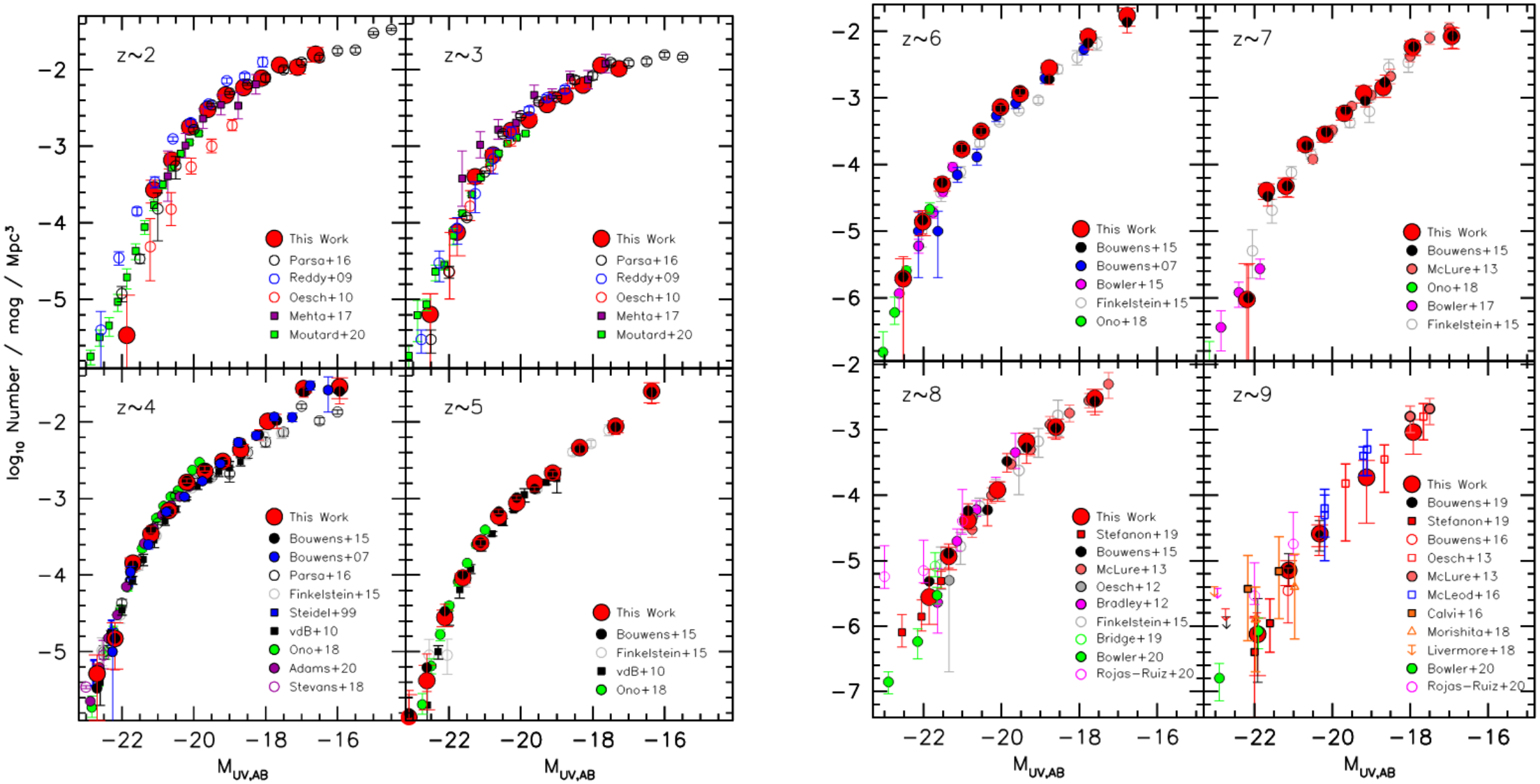




Can now measure large samples across many redshifts self-consistently

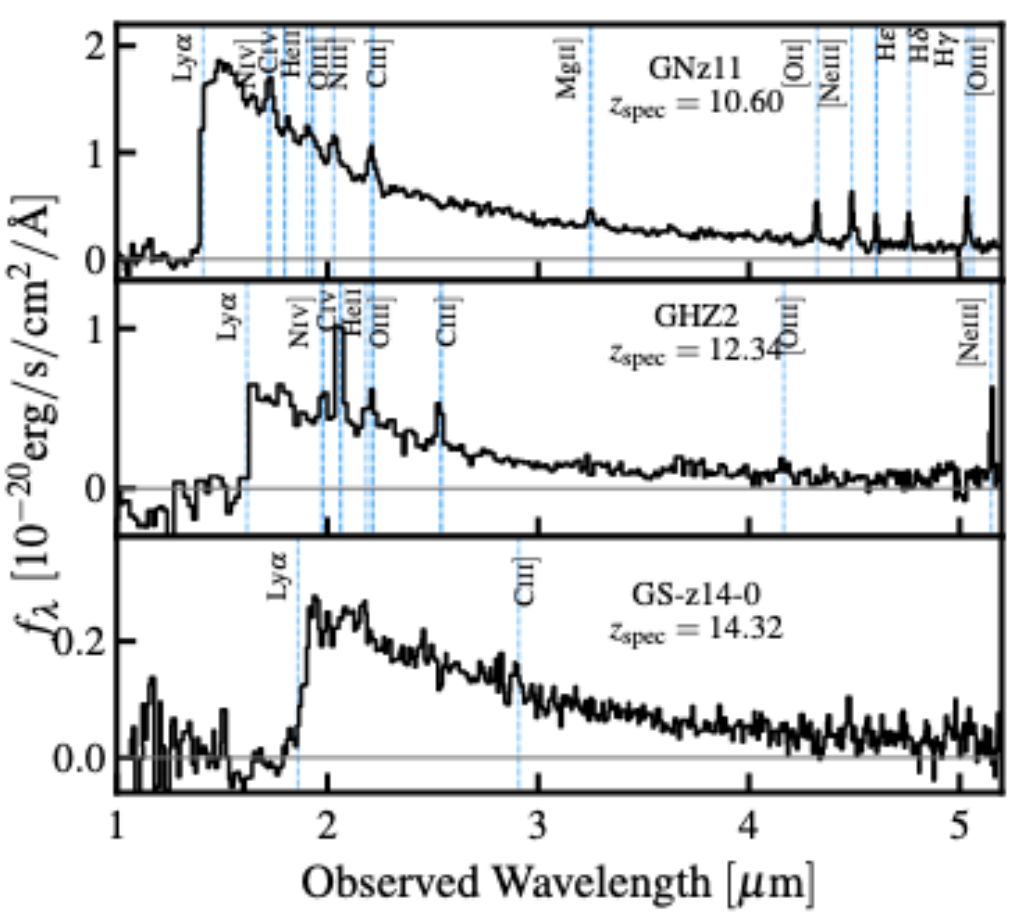
Bouwens et al 2021

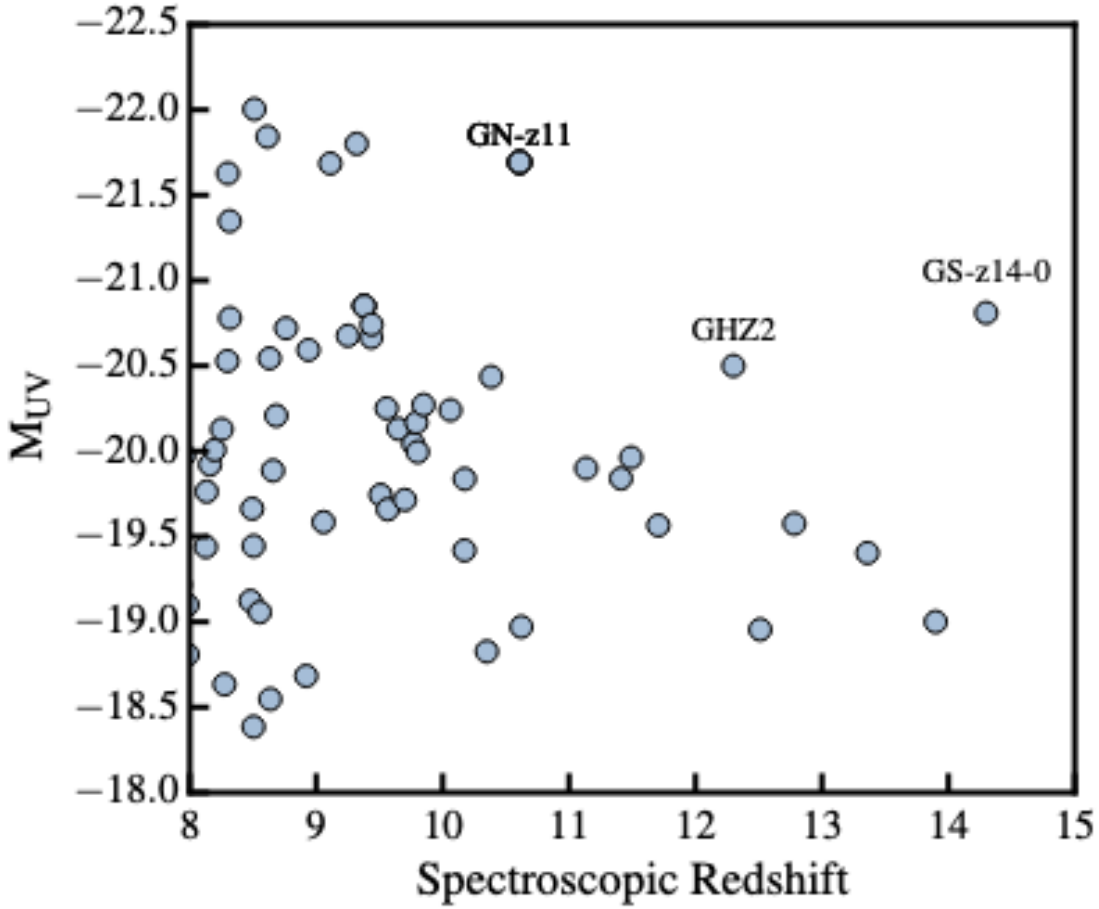
High *z* galaxies



Bouwens et al 2021

High *z* galaxies





Bouwens et al 2021

High redshift galaxies (8 < z < 20)

HST

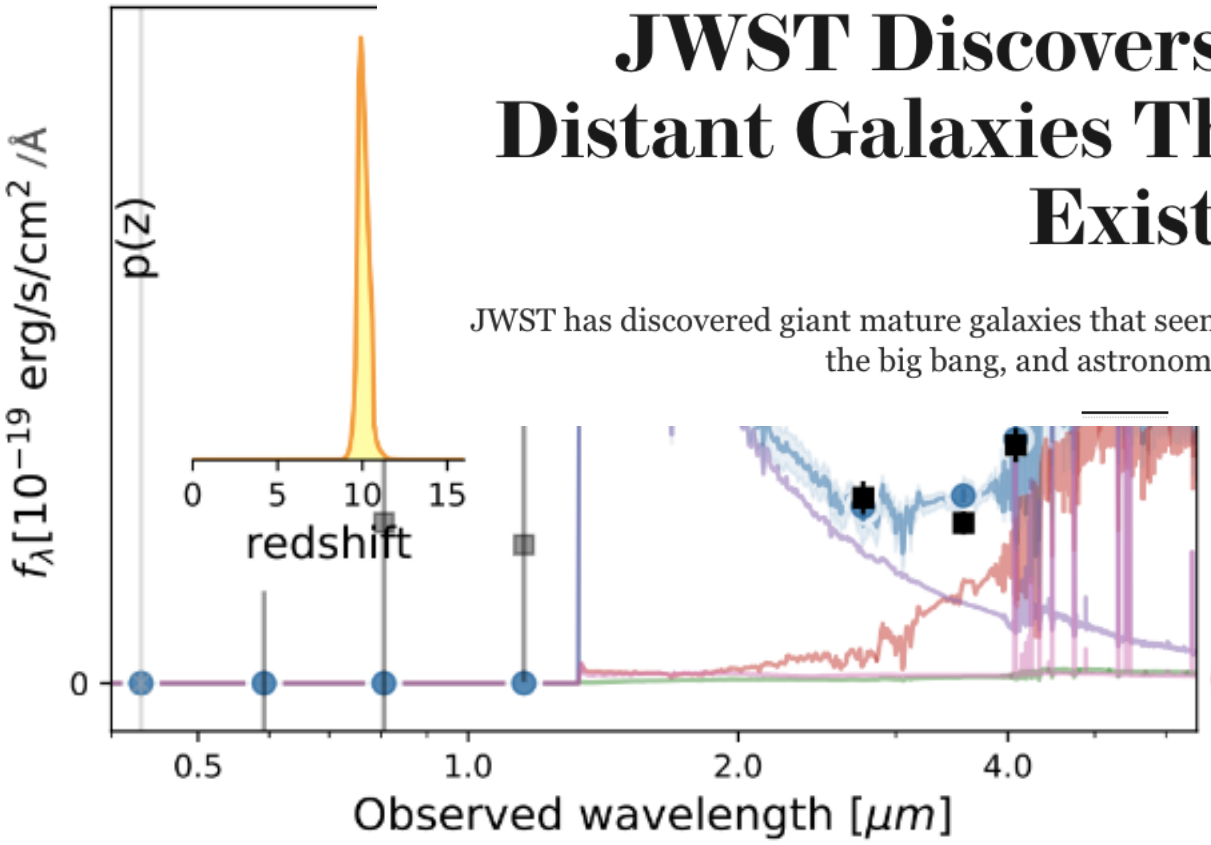
JWST

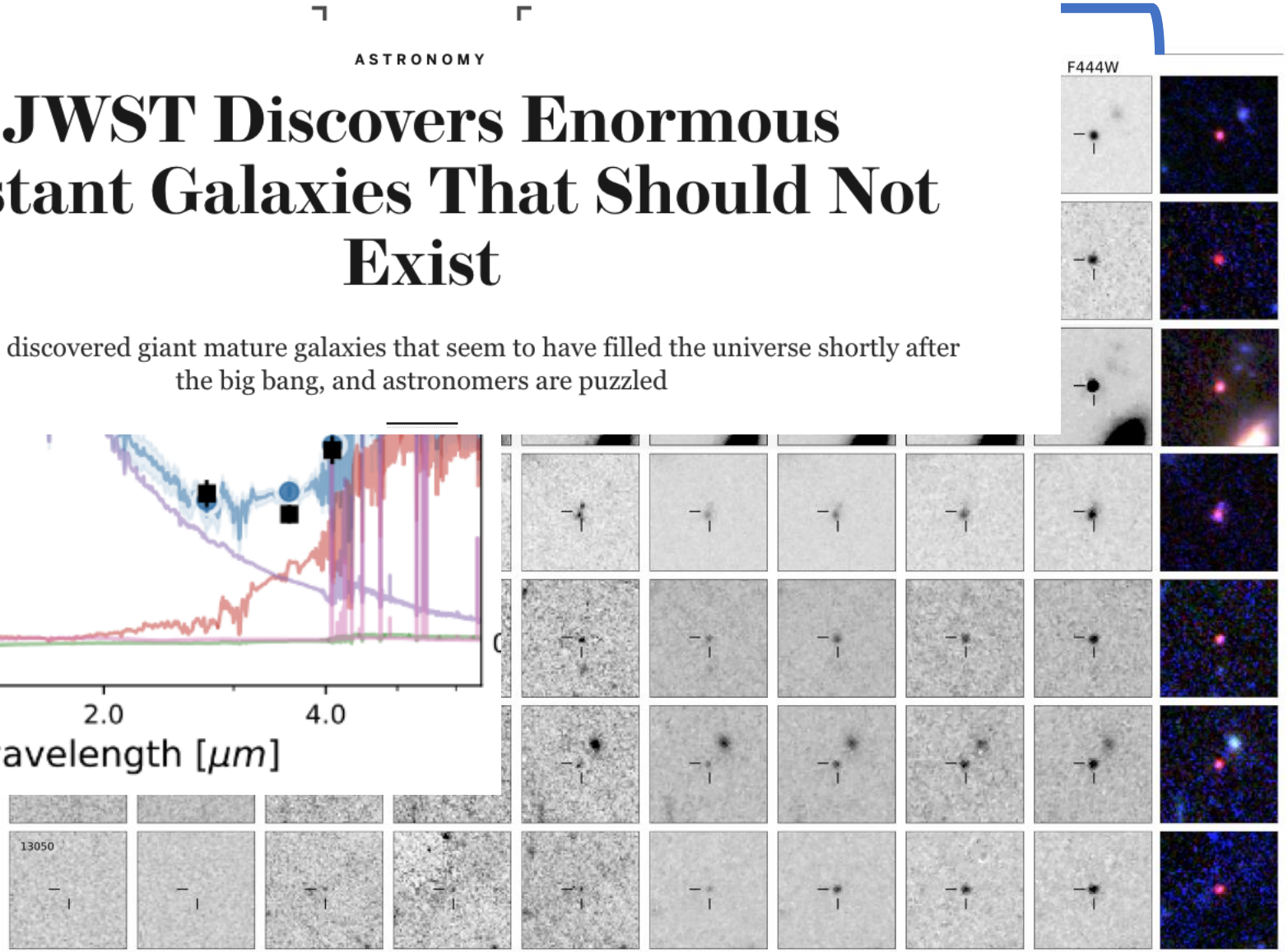
FUTURE

ASTRONOMY

JWST Discovers Enormous Distant Galaxies That Should Not Exist

JWST has discovered giant mature galaxies that seem to have filled the universe shortly after the big bang, and astronomers are puzzled





Labbe et al. 2022

Why are these galaxies thought to be 'too massive to exist'?

Press-Schechter model:

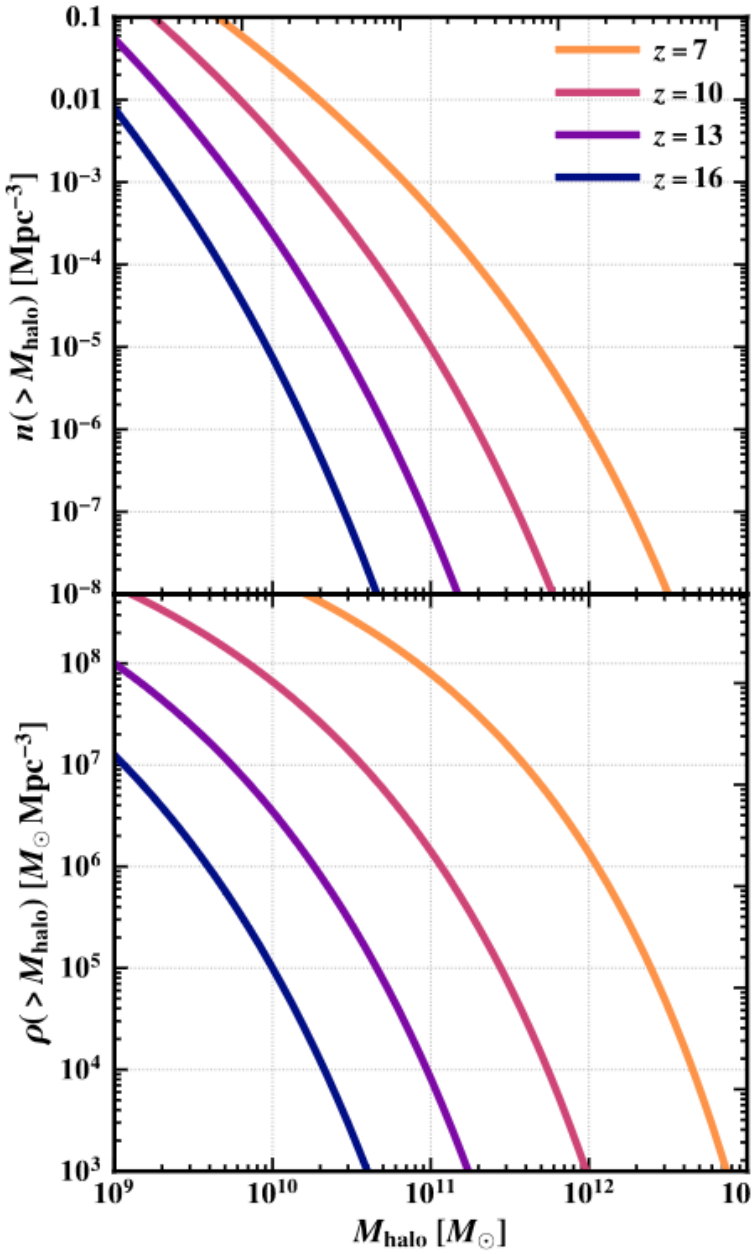
$$\frac{dn}{dM}(M,t) = \frac{2\bar{\rho}\delta_c}{M^2}\frac{1}{\sqrt{2\pi}\sigma_M}\left|\frac{dln\sigma_M}{dlnM}\right|\exp\left(\frac{-\delta_c^2}{2\sigma_M^2}\right)$$

Calculate the number of halos more massive than a particular halo limit (at a fixed redshift):

$$n(>M_{\rm halo},z) = \int_{M_{\rm halo}}^{\infty} dM\,\frac{dn(M,z)}{dM}$$

Or the total mass density in halos more massive than a particular halo limit:

$$\rho(>M_{\rm halo},z) = \int_{M_{\rm halo}}^{\infty} dM\,M\,\frac{dn(M,z)}{dM}\,.$$



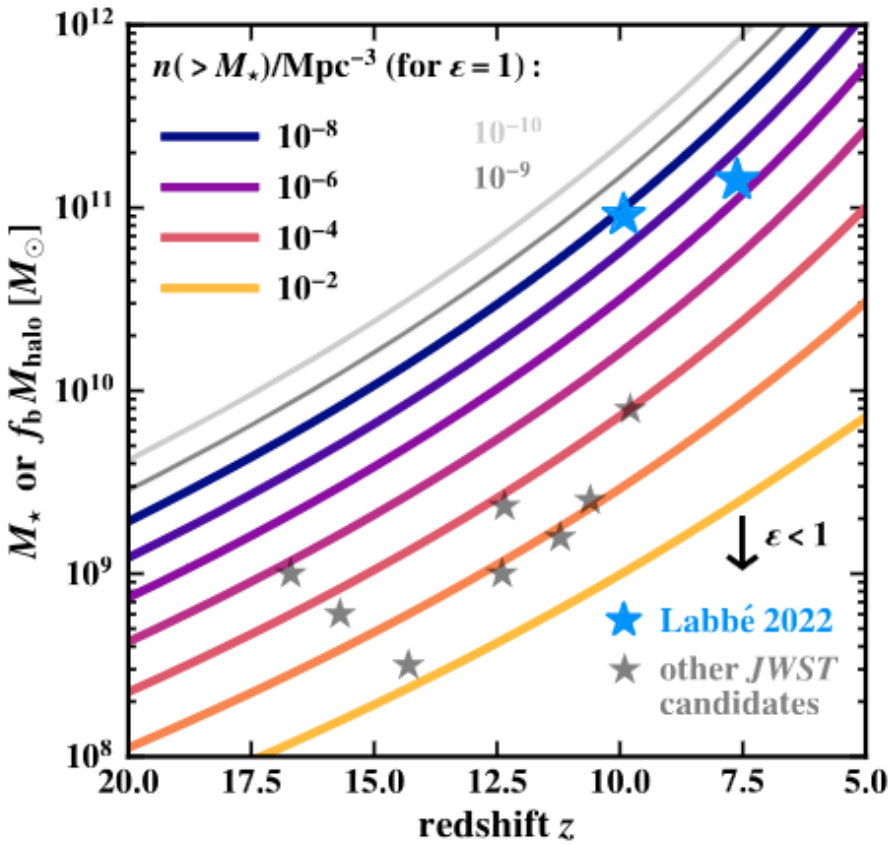
Why are these galaxies thought to be 'too massive to exist'?

These are for 'total mass', how much baryonic material is available:

$$M_\star = \epsilon f_b M_{halo}, \text{ with } \epsilon \leq 1$$

Roughly, it's unlikely in a survey of volume V to find a galaxy whose stellar mass corresponds to n(> M*,max) = V^{-1}

The volume of JWST surveys is $\sim V^{-1} = 10^{-5}\ Mpc^{-3}$



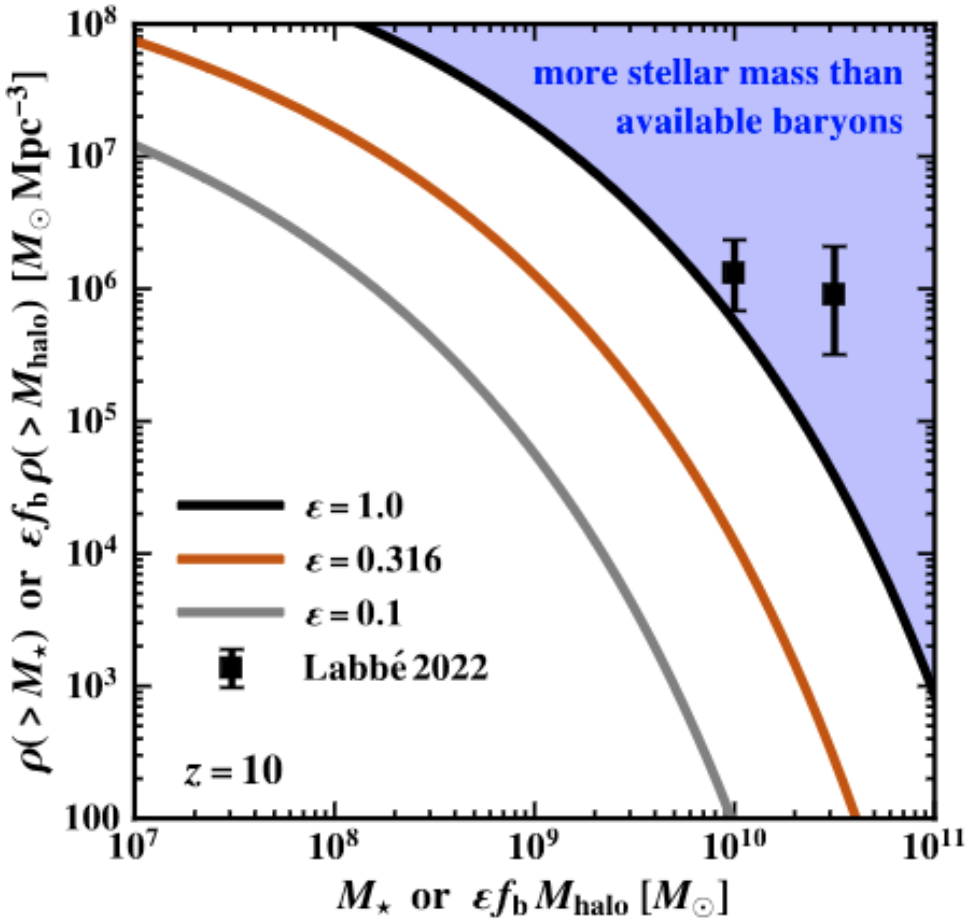
Why are these galaxies thought to be 'too massive to exist'?

These are for 'total mass', how much baryonic material is available:

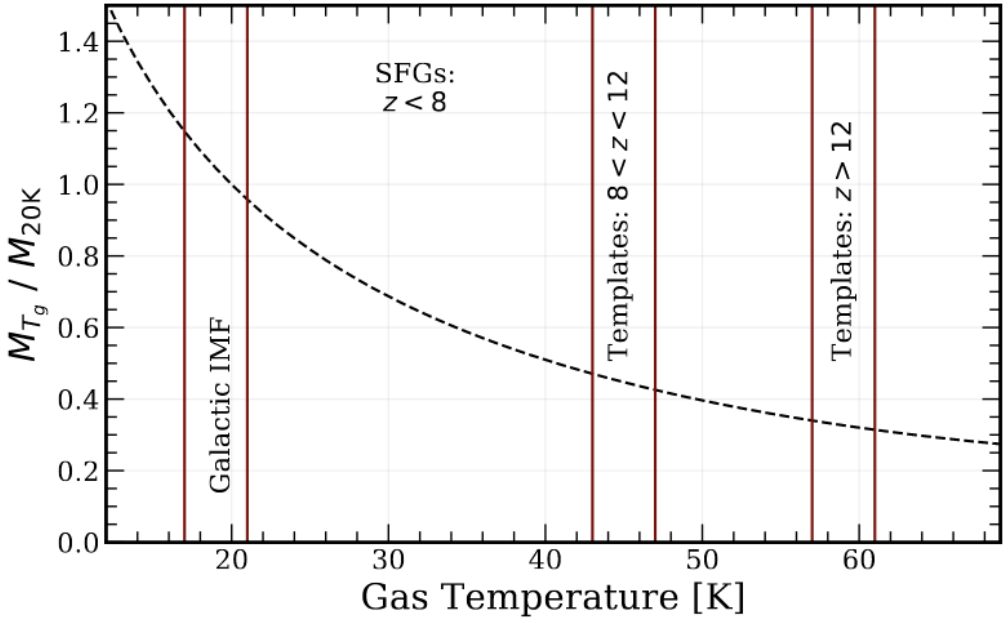
$$M_\star = \epsilon f_b M_{halo}, \text{ with } \epsilon \leq 1$$

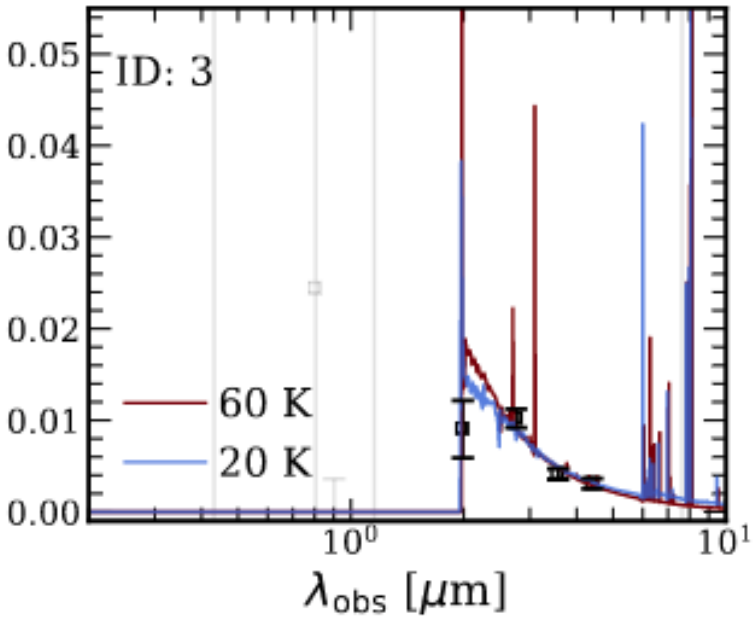
Roughly, it's unlikely in a survey of volume V to find a galaxy whose stellar mass corresponds to n(> M*,max) = V^{-1}

The volume of JWST surveys is $\sim V^{-1} = 10^{-5}\ Mpc^{-3}$



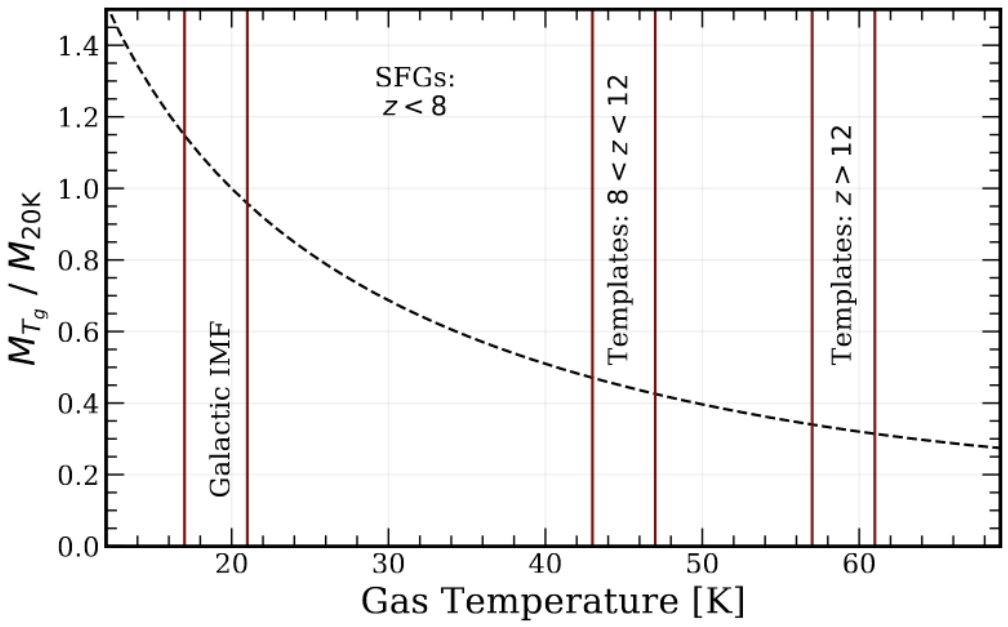
Requirement for ‘new’ physics?

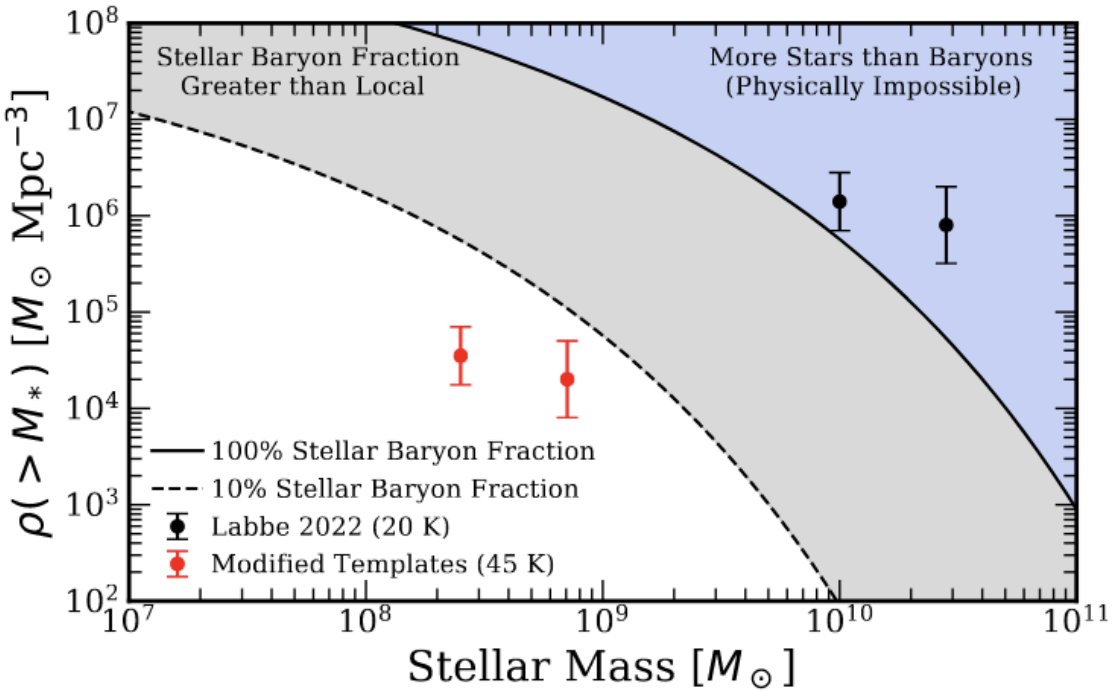




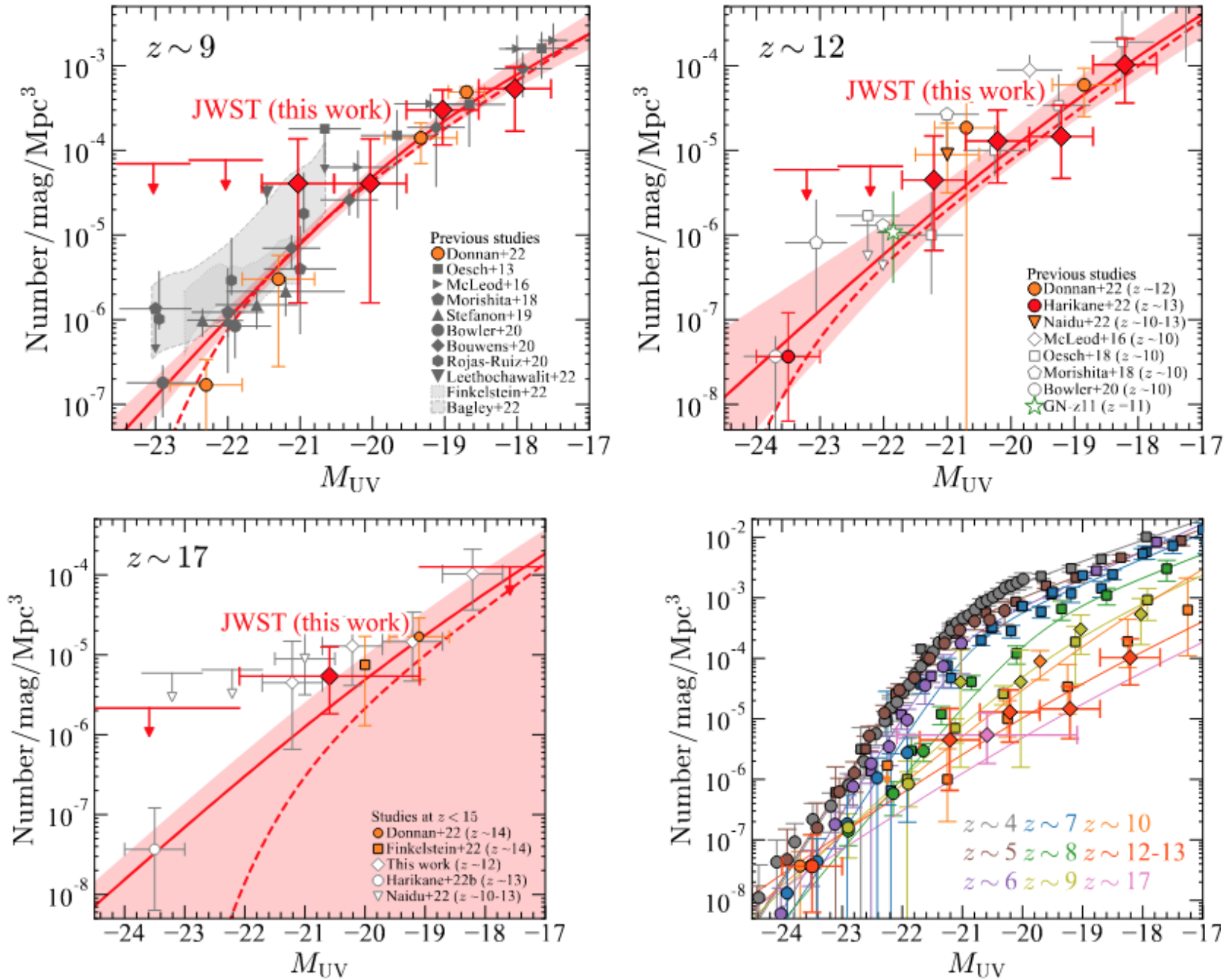
Steinhardt et al. 2022

Requirement for ‘new’ physics?



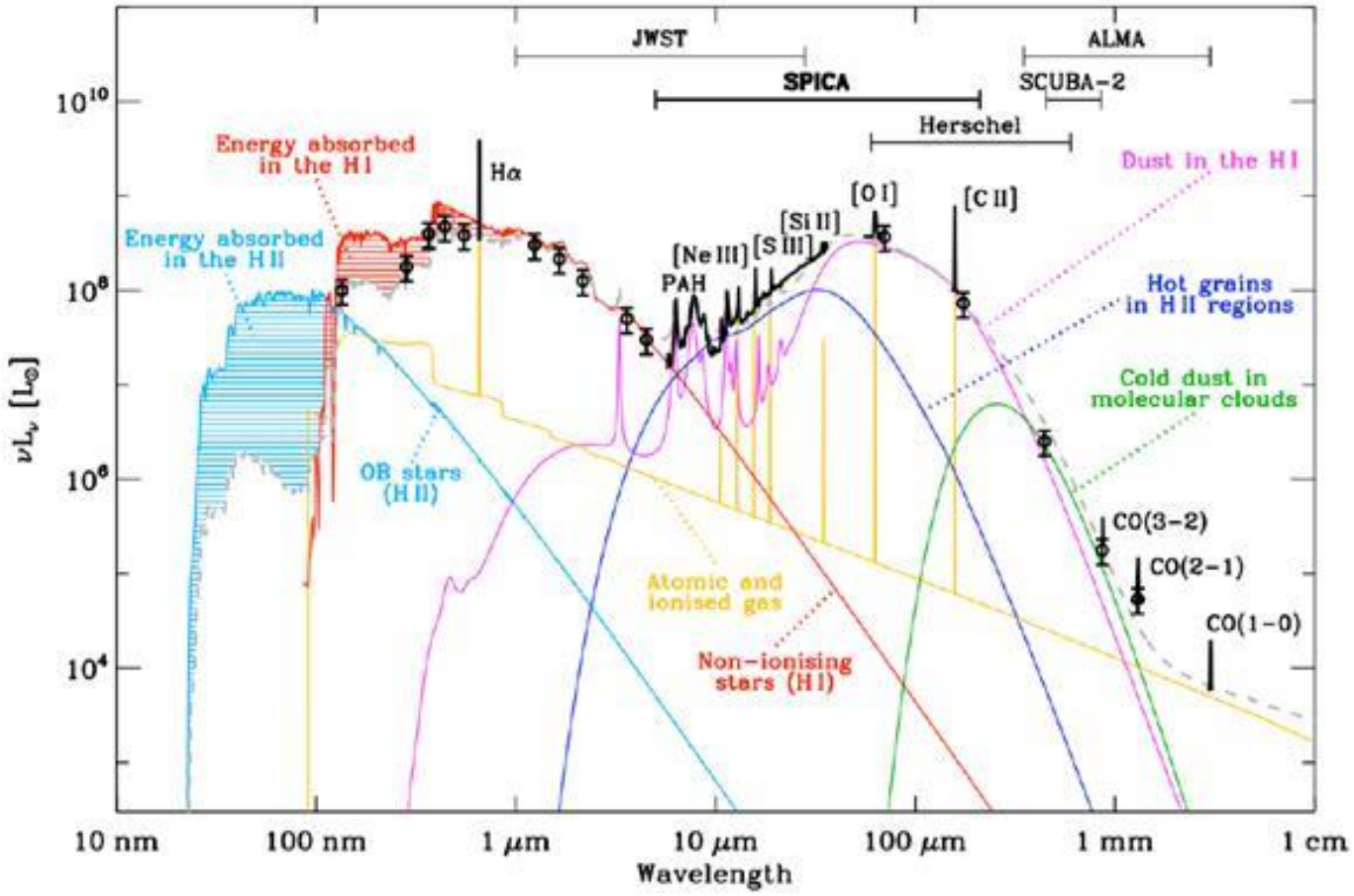


Steinhardt et al. 2022



Harikane et al. 2022

The spectral energy distribution of galaxies

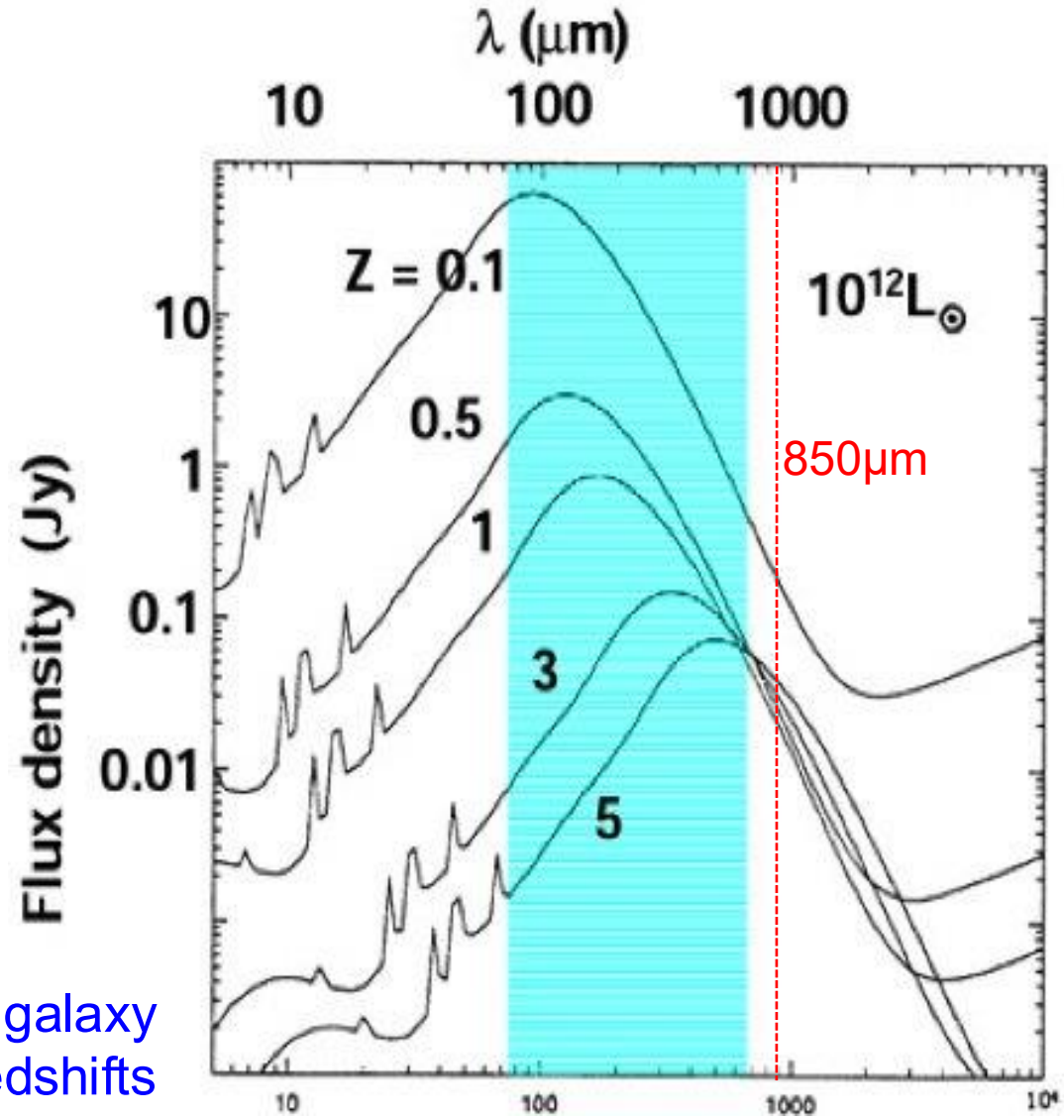


Need to exploit the SED of galaxies to find high redshift ones.

(Galliano et al. 2008)

Submillimetre galaxies

With increasing redshift, the bolometric flux from a galaxy is reduced as d_L^{-2}. The spectrum is also shifted towards the red. In cases where the spectrum is rising steeply towards shorter wavelengths, this redshifting will cause a *brighter* portion of the spectrum to be sampled at a given observed wavelength – offsetting the d^{-2} dimming.



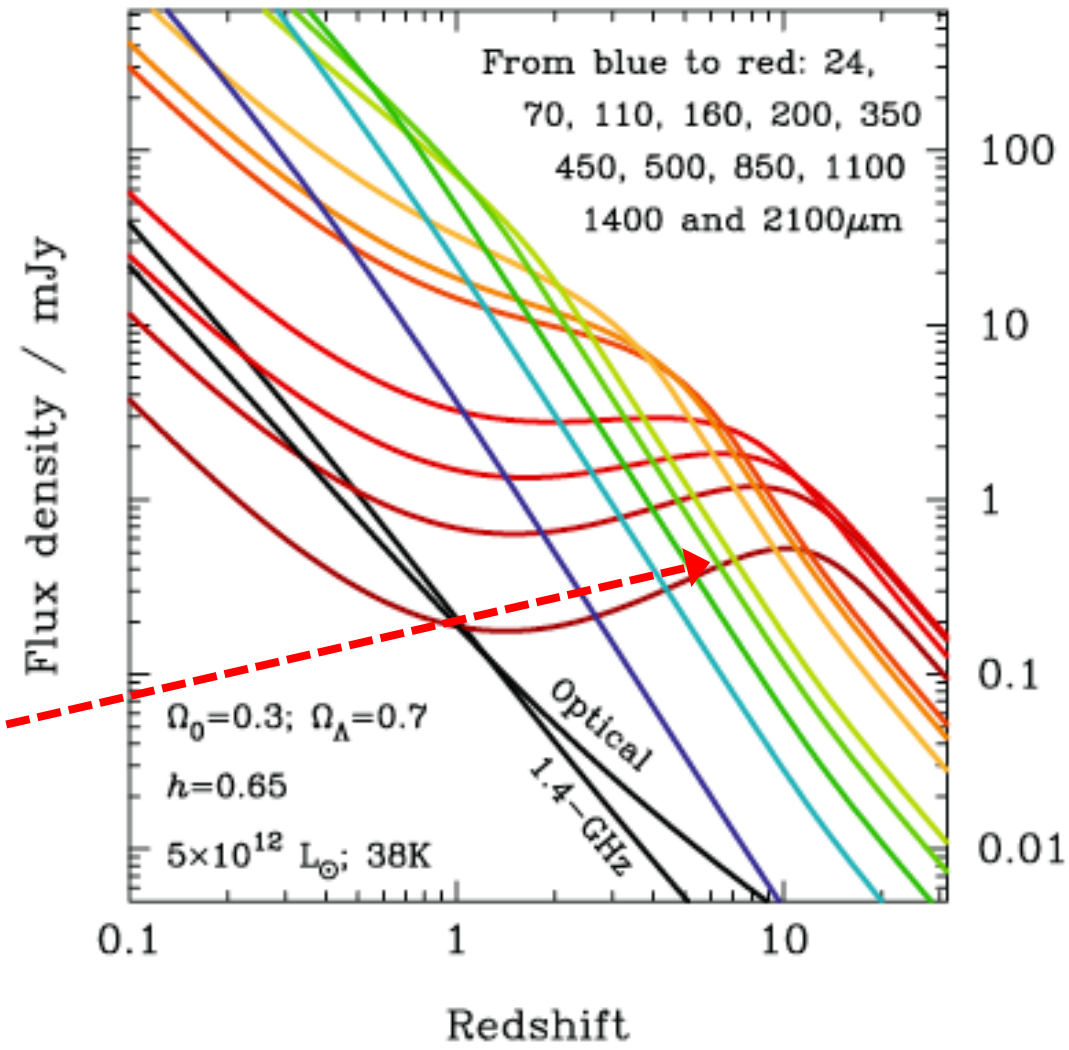
Spectrum of star-forming galaxy over a range of redshifts

Submillimetre galaxies

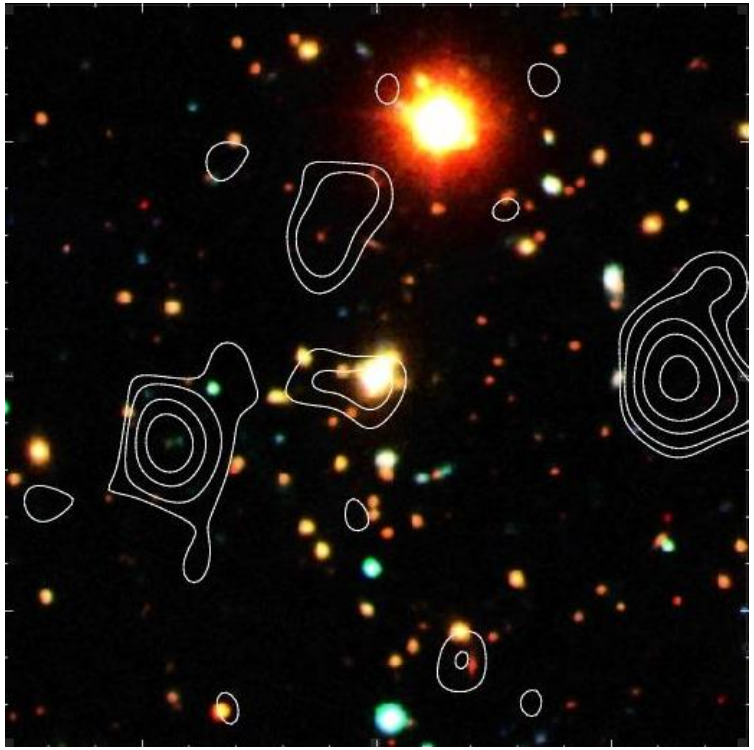
More formally, K-correction is the dimming due to the (1+z) shifting of the wavelength band (and its width) for a filter with response S(λ).

$$k(z) = (1+z)\frac{\int F(\lambda)S(\lambda)d\lambda}{\int F(\lambda/1+z)S(\lambda)d\lambda}$$

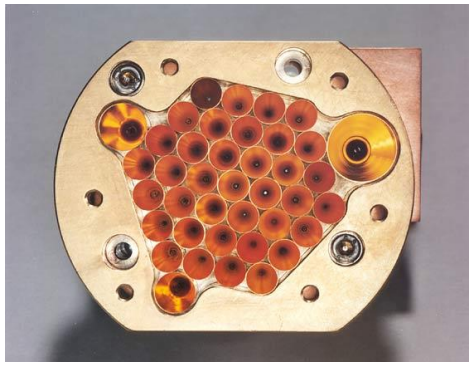
In the Rayleigh-Jeans (long λ) tail of the dust blackbody spectrum, galaxies get ***brighter*** as they are redshifted to greater distance!



Submillimetre galaxies



15m JCMT



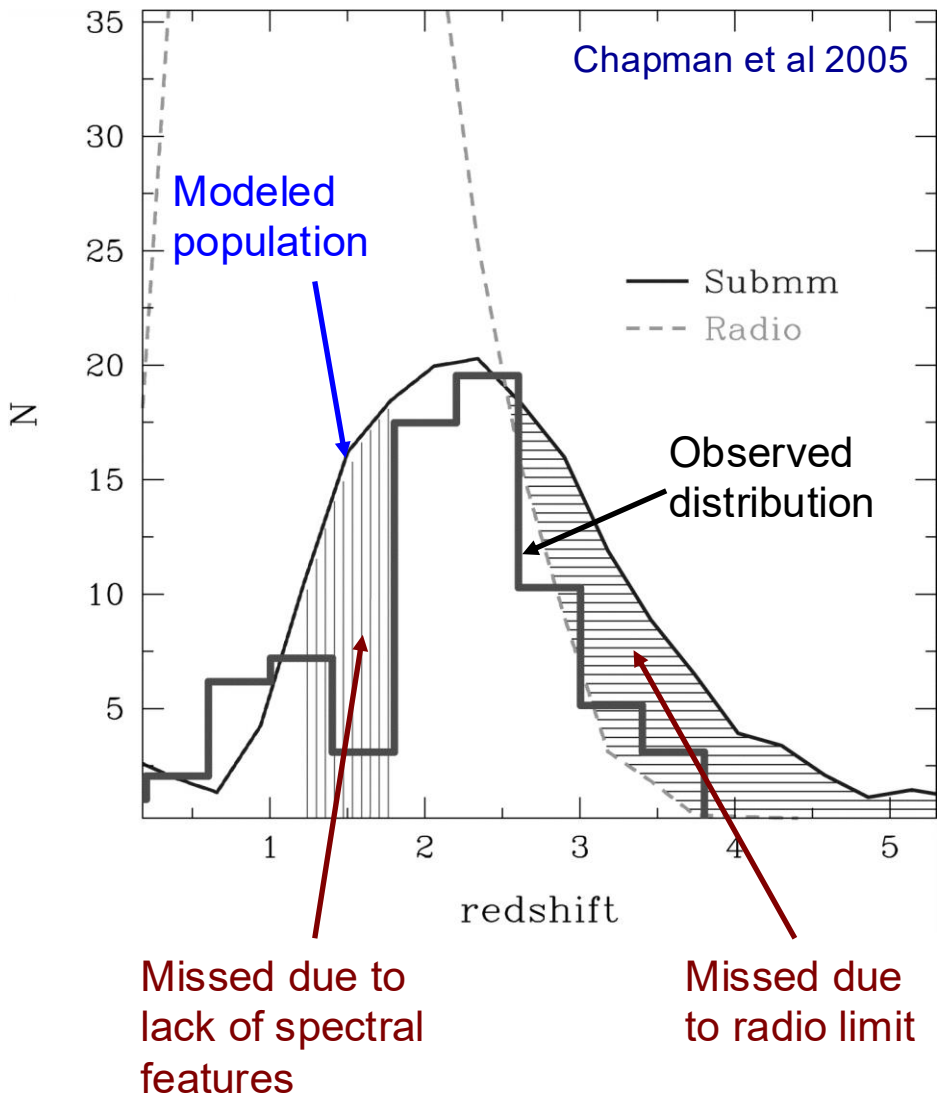
SCUBA array

SCUBA: 850μm array detects dusty star forming sources:
- behind lensing clusters (Smail et al 1997)
- in blind surveys (Hughes et al 1998)

Source density implies 1000-fold excess over no evolution model based on density of local IRAS sources:

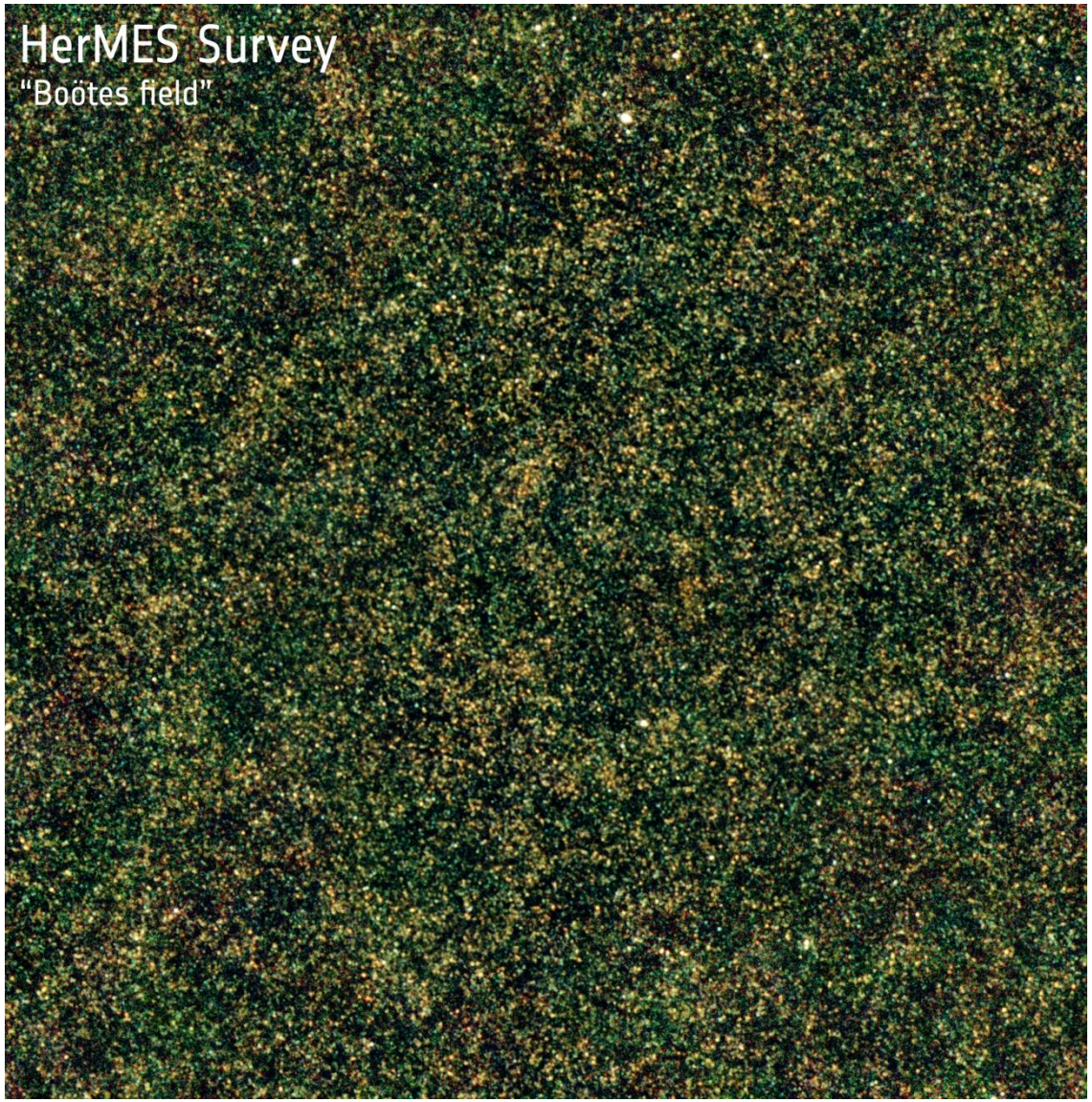
Key questions: what are their typical redshift, luminosity and SFR?

Submillimetre galaxies



- ~50% completeness for submm sources with 850 μm flux>5 mJy, plus radio counterparts and deep optical spectroscopy
- Most sub-mm sources have z < 4
- Peak at z ≈ 2.3 – comparable to that for AGN
- Bolometric luminosities are ~ 10^{13} $L_\odot$, and inferred SFRs ~1000 $M_\odot$ yr^{-1}.
- Although ρ(LBG) ~ 10 ρ(SCUBA), luminosity/SF densities are comparable

Submillimetre galaxies



- ~50% completeness for submm sources with 850 μm flux>5 mJy, plus radio counterparts and deep optical spectroscopy
- Most sub-mm sources have $z < 4$
- Peak at $z \approx 2.3$ – comparable to that for AGN
- Bolometric luminosities are ~ 10^{13} $L_\odot$, and inferred SFRs ~1000 $M_\odot$ yr^{-1}.
- Although ρ(LBG) ~ 10 ρ(SCUBA), luminosity/SF densities are comparable